

# MATHEMATICS OF GENERATIVE MODELS

IZ TECÚM

ABSTRACT. SLMath notes.

## CONTENTS

|                                      |    |
|--------------------------------------|----|
| <b>Part 1. Lecture A<sup>1</sup></b> | 2  |
| 1. Measure Transport                 | 2  |
| 1.1. Flows                           | 2  |
| 1.2. Interpolants                    | 4  |
| 2. Diffusive Interpolants            | 7  |
| 2.1. Tunable Noise                   | 7  |
| 2.2. Brownian Paths                  | 8  |
| <b>Part 2. Lecture B<sup>2</sup></b> | 10 |
| 3. Langevin Sampling                 | 10 |
| 3.1. Kernels                         | 10 |
| 3.2. Dissipation                     | 11 |
| 3.3. Discretization                  | 13 |
| 4. Score Diffusions                  | 14 |
| 4.1. Reversal                        | 14 |
| 4.2. Localization                    | 15 |
| 4.3. Bookkeeping                     | 17 |
| <b>Part 3. Exercises</b>             | 19 |
| 5. Exercises                         | 19 |
| 6. Homework                          | 20 |
| References                           | 23 |

---

*Date:* September 4, 2026.

<sup>1</sup>Based on lectures by Jianfeng Lu [15] .

<sup>2</sup>Based on lectures by Eric Vanden-Eijnden [22] .

## Part 1. Lecture A<sup>3</sup>

### 1. MEASURE TRANSPORT

Throughout,  $\mu_0$  is a base law on  $\mathbb{R}^d$  that can be sampled and evaluated,  $\mu_1$  is a target known only through samples, and  $\mu_t := \text{Law}(\mathbf{X}_t)$  is the law transported by a velocity field  $\mathbf{b}_t$ ; hatted objects  $\hat{\mathbf{b}}, \hat{p}$  are learned. Boldface denotes vectors and vector fields,  $\mathbf{N}(\mathbf{m}, C)$  is Gaussian,  $\Pi(\mu_0, \mu_1)$  is the set of couplings, densities are  $p_t$ , scores are  $\mathbf{s}_t = \nabla \log p_t$ , and  $\tau \sim \text{Unif}[0, 1]$  is independent of all interpolant data [9, 18, 2].

#### 1.1. Flows.

**Definition 1.1.1** (Time-dependent flow and transported law). Let  $\mathbf{b} : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ , let  $\dot{\mathbf{X}}_t = \mathbf{b}_t(\mathbf{X}_t)$  with  $\mathbf{X}_0 \sim \mu_0$ , and let  $\Phi_{t,0}$  be the flow map. The transported law is

$$\mu_t := \text{Law}(\mathbf{X}_t) = (\Phi_{t,0})_{\#} \mu_0, \quad \mu_t(A) = \mu_0(\Phi_{t,0}^{-1}(A)) \quad (A \subset \mathbb{R}^d \text{ Borel}).$$

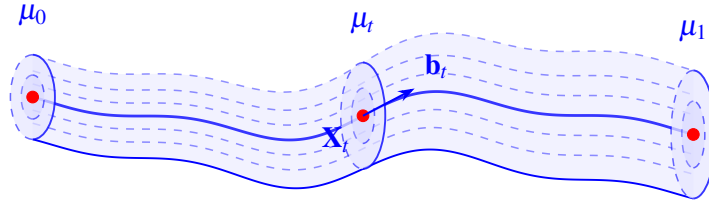


FIGURE 1. Characteristics of  $\mathbf{b}_t$  transport  $\mu_0$  through  $\mu_t$  to  $\mu_1$ .

**Proposition 1.1.2** (Weak continuity equation). If  $\mathbf{b}$  and the flow of (1.1.1) are regular enough to differentiate under the expectation, then

$$\frac{d}{dt} \int_{\mathbb{R}^d} \varphi d\mu_t = \int_{\mathbb{R}^d} \nabla \varphi \cdot \mathbf{b}_t d\mu_t \quad (\varphi \in C_c^1(\mathbb{R}^d)),$$

equivalently, in the sense of distributions,

$$(1.1.1) \quad \partial_t \mu_t + \nabla \cdot (\mathbf{b}_t \mu_t) = 0.$$

*Proof.*

$$\begin{aligned} \int_{\mathbb{R}^d} \varphi d\mu_t &\stackrel{\mu_t = \text{Law}(\mathbf{X}_t)}{\iff} \mathbb{E}[\varphi(\mathbf{X}_t)] \stackrel{\text{differentiate under } \mathbb{E}}{\implies} \mathbb{E}[\nabla \varphi(\mathbf{X}_t) \cdot \dot{\mathbf{X}}_t] \stackrel{1.1.1}{\implies} \int_{\mathbb{R}^d} \nabla \varphi \cdot \mathbf{b}_t d\mu_t, \\ \int_0^1 \int_{\mathbb{R}^d} (\partial_t \psi + \mathbf{b}_t \cdot \nabla \psi) d\mu_t dt &= 0 \quad (\psi \in C_c^\infty) \stackrel{\text{test in space and time}}{\iff} (1.1.1). \end{aligned}$$

□

**Corollary 1.1.3** (Density form). If  $\mu_t = p_t d\mathbf{x}$ , then

$$\partial_t p_t + \nabla \cdot (\mathbf{b}_t p_t) = 0, \quad \partial_t p_t + \mathbf{b}_t \cdot \nabla p_t + (\nabla \cdot \mathbf{b}_t) p_t = 0 \quad \text{for smooth } p_t, \mathbf{b}_t.$$

*Proof.*

$$\begin{aligned} \int \varphi \partial_t p_t d\mathbf{x} &\stackrel{1.1.2}{\implies} \int \nabla \varphi \cdot \mathbf{b}_t p_t d\mathbf{x} \stackrel{\text{integrate by parts}}{\implies} - \int \varphi \nabla \cdot (\mathbf{b}_t p_t) d\mathbf{x} \stackrel{\varphi \in C_c^1 \text{ arbitrary}}{\implies} \partial_t p_t + \nabla \cdot (\mathbf{b}_t p_t) = 0 \\ &\stackrel{\text{Leibniz on } \nabla \cdot (\mathbf{b}_t p_t)}{\iff} \partial_t p_t + \mathbf{b}_t \cdot \nabla p_t + (\nabla \cdot \mathbf{b}_t) p_t = 0. \end{aligned}$$

□

<sup>3</sup>Based on lectures by Jianfeng Lu [15].

**Example 1.1.4** (Gaussian dilation). For  $\dot{\mathbf{X}}_t = \lambda \mathbf{X}_t$  and  $\mathbf{X}_0 \sim \mathcal{N}(\mathbf{0}, C_0)$ ,

$$\begin{aligned} \mathbf{X}_t &= e^{\lambda t} \mathbf{X}_0, & \mu_t &= \mathcal{N}(\mathbf{0}, C_t), & C_t &= e^{2\lambda t} C_0, & \partial_t \log \det(C_t) &= 2\lambda d, & \partial_t C_t^{-1} &= -2\lambda C_t^{-1}, \\ \partial_t p_t &= (-d\lambda + \lambda \mathbf{x}^\top \cdot C_t^{-1} \cdot \mathbf{x}) p_t, & \nabla \cdot (\lambda \mathbf{x} p_t) &= (\lambda d - \lambda \mathbf{x}^\top \cdot C_t^{-1} \cdot \mathbf{x}) p_t \xrightarrow{1.1.3} \partial_t p_t + \nabla \cdot (\lambda \mathbf{x} p_t) = 0. \end{aligned}$$

**Proposition 1.1.5** (Instantaneous change of variables). *Wherever  $p_t(\mathbf{X}_t) > 0$ ,*

$$\frac{d}{dt} \log p_t(\mathbf{X}_t) = -(\nabla \cdot \mathbf{b}_t)(\mathbf{X}_t), \quad \log p_1(\mathbf{X}_1) = \log p_0(\mathbf{X}_0) - \int_0^1 (\nabla \cdot \mathbf{b}_t)(\mathbf{X}_t) dt.$$

*Proof.*

$$\begin{aligned} \frac{d}{dt} p_t(\mathbf{X}_t) &= \partial_t p_t(\mathbf{X}_t) + \nabla p_t(\mathbf{X}_t) \cdot \dot{\mathbf{X}}_t \xrightarrow{1.1.3, 1.1.1} -\nabla \cdot (\mathbf{b}_t p_t) + \mathbf{b}_t \cdot \nabla p_t \xLeftrightarrow{\text{Leibniz}} -(\nabla \cdot \mathbf{b}_t) p_t(\mathbf{X}_t), \\ \frac{d}{dt} \log p_t(\mathbf{X}_t) &= -(\nabla \cdot \mathbf{b}_t)(\mathbf{X}_t) \xrightarrow{\text{integrate over } [0, 1]} \log p_1(\mathbf{X}_1) = \log p_0(\mathbf{X}_0) - \int_0^1 (\nabla \cdot \mathbf{b}_t)(\mathbf{X}_t) dt. \end{aligned}$$

□

**Definition 1.1.6** (Continuous normalizing flow likelihood). Let  $p_0$  be evaluable and  $\widehat{\mathbf{b}}$  parameterize an invertible flow. Solving  $\dot{\mathbf{X}}_t = \widehat{\mathbf{b}}_t(\mathbf{X}_t)$  backward from  $\mathbf{X}_1 = \mathbf{x}_1$ ,

$$\log \widehat{p}_1(\mathbf{x}_1) \xLeftrightarrow{1.1.5} \log p_0(\mathbf{X}_0) - \int_0^1 (\nabla \cdot \widehat{\mathbf{b}}_t)(\mathbf{X}_t) dt, \quad \text{CE}(\widehat{\mathbf{b}}) := -\mathbb{E}_{\mathbf{X}_1 \sim \mu_1} \log \widehat{p}_1(\mathbf{X}_1).$$

**Definition 1.1.7** (Quadratic Monge transport). A measurable  $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$  transports  $\mu_0$  to  $\mu_1$  when  $T_\# \mu_0 = \mu_1$ ; its quadratic cost is

$$\mathcal{C}(T) := \int_{\mathbb{R}^d} \|T(\mathbf{x}) - \mathbf{x}\|^2 \mu_0(d\mathbf{x}).$$

**Proposition 1.1.8** (Kinetic action controls endpoint displacement). *For  $\mathbf{X}_t$  as in (1.1.1) on  $[0, 1]$ ,*

$$\mathbb{E} \|\mathbf{X}_1 - \mathbf{X}_0\|^2 \leq \int_0^1 \int_{\mathbb{R}^d} \|\mathbf{b}_t(\mathbf{x})\|^2 \mu_t(d\mathbf{x}) dt.$$

*Proof.*

$$\begin{aligned} \mathbf{X}_1 - \mathbf{X}_0 &= \int_0^1 \mathbf{b}_t(\mathbf{X}_t) dt \xrightarrow{\text{Cauchy-Schwarz in } t} \|\mathbf{X}_1 - \mathbf{X}_0\|^2 \leq \int_0^1 \|\mathbf{b}_t(\mathbf{X}_t)\|^2 dt \\ &\xrightarrow{\mathbb{E}, \text{Fubini}, \mu_t = \text{Law}(\mathbf{X}_t)} \mathbb{E} \|\mathbf{X}_1 - \mathbf{X}_0\|^2 \leq \int_0^1 \int \|\mathbf{b}_t\|^2 d\mu_t dt \quad (2). \end{aligned}$$

□

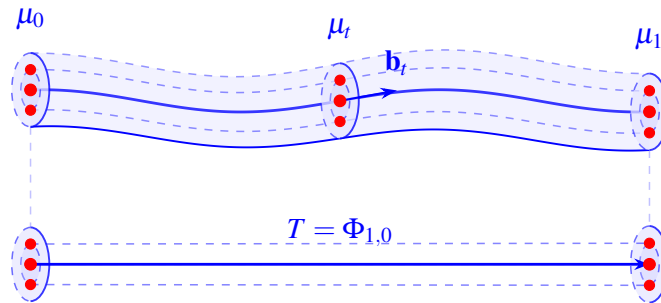


FIGURE 2. The action of  $(\mu_t, \mathbf{b}_t)$  bounds the quadratic cost of  $T = \Phi_{1,0}$ .

## 1.2. Interpolants.

**Definition 1.2.1** (Coupling and affine stochastic interpolant). Let  $(\mathbf{X}_0, \mathbf{X}_1) \sim \pi \in \Pi(\mu_0, \mu_1)$  and let  $\alpha, \beta \in C^1([0, 1])$  satisfy  $\alpha_0 = \beta_1 = 1$ ,  $\alpha_1 = \beta_0 = 0$ . Then

$$\mathbf{I}_t := \alpha_t \mathbf{X}_0 + \beta_t \mathbf{X}_1, \quad \dot{\mathbf{I}}_t = \dot{\alpha}_t \mathbf{X}_0 + \dot{\beta}_t \mathbf{X}_1, \quad \mu_t := \text{Law}(\mathbf{I}_t).$$

**Definition 1.2.2** (Interpolant velocity and denoisers). Wherever the conditional expectations exist,

$$\mathbf{b}_t(\mathbf{x}) := \mathbb{E}[\dot{\mathbf{I}}_t \mid \mathbf{I}_t = \mathbf{x}], \quad \eta_t^0(\mathbf{x}) := \mathbb{E}[\mathbf{X}_0 \mid \mathbf{I}_t = \mathbf{x}], \quad \eta_t^1(\mathbf{x}) := \mathbb{E}[\mathbf{X}_1 \mid \mathbf{I}_t = \mathbf{x}],$$

$$\mathbf{b}_t \xLeftrightarrow{1.2.1} \dot{\alpha}_t \eta_t^0 + \dot{\beta}_t \eta_t^1, \quad \mathbf{x} \xLeftrightarrow{1.2.1} \alpha_t \eta_t^0(\mathbf{x}) + \beta_t \eta_t^1(\mathbf{x}).$$

**Theorem 1.2.3** (Probability flow of an affine interpolant). *The pair  $(\mu_t, \mathbf{b}_t)$  of (1.2.2) satisfies (1.1.1). If  $\dot{\mathbf{Y}}_t = \mathbf{b}_t(\mathbf{Y}_t)$ ,  $\mathbf{Y}_0 \sim \mu_0$ , is well posed and the continuity equation is unique in the relevant class, then*

$$\text{Law}(\mathbf{Y}_t) = \mu_t \quad (0 \leq t \leq 1), \quad \text{Law}(\mathbf{Y}_1) = \mu_1.$$

*Proof.*

$$\begin{aligned} \frac{d}{dt} \mathbb{E}[\varphi(\mathbf{I}_t)] &= \mathbb{E}[\nabla \varphi(\mathbf{I}_t) \cdot \dot{\mathbf{I}}_t] \xrightarrow{\text{tower over } \sigma(\mathbf{I}_t)} \mathbb{E}[\nabla \varphi(\mathbf{I}_t) \cdot \mathbb{E}[\dot{\mathbf{I}}_t \mid \mathbf{I}_t]] \xrightarrow{1.2.2} \int \nabla \varphi \cdot \mathbf{b}_t d\mu_t \\ &\xLeftrightarrow{1.1.2} (1.1.1) \xrightarrow{\text{Law}(\mathbf{Y}_0) = \mu_0, \text{ uniqueness}} \text{Law}(\mathbf{Y}_t) = \mu_t \quad (3). \end{aligned}$$

□

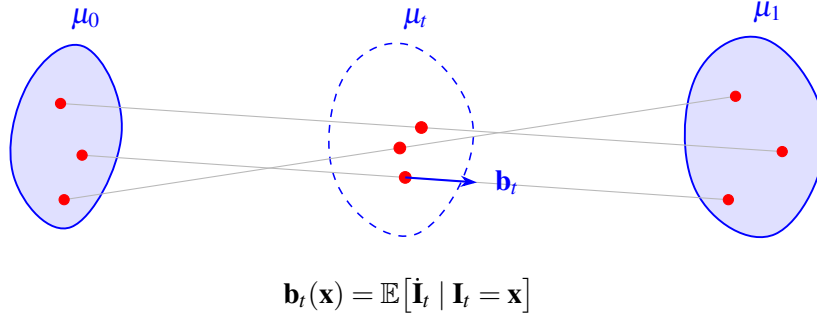


FIGURE 3. The coupling draws random chords, the interpolant rides them, and conditional averaging of the chord velocities produces the Eulerian field  $\mathbf{b}_t$ .

**Example 1.2.4** (Independent linear interpolation). For  $\mathbf{X}_0 \perp \mathbf{X}_1$ ,  $\alpha_t = 1 - t$ ,  $\beta_t = t$ ,

$$\mathbf{I}_t = (1 - t)\mathbf{X}_0 + t\mathbf{X}_1, \quad \dot{\mathbf{I}}_t = \mathbf{X}_1 - \mathbf{X}_0 \xrightarrow{1.2.2} \mathbf{b}_t(\mathbf{x}) = \mathbb{E}[\mathbf{X}_1 - \mathbf{X}_0 \mid \mathbf{I}_t = \mathbf{x}].$$

**Lemma 1.2.5** (Conditional expectation as orthogonal projection). *For  $\mathbf{U} \in L^2$ , a random vector  $\mathbf{V}$ , and measurable  $\mathbf{f}$  with  $\mathbf{f}(\mathbf{V}) \in L^2$ ,*

$$\mathbb{E}\|\mathbf{f}(\mathbf{V}) - \mathbf{U}\|^2 = \mathbb{E}\|\mathbf{f}(\mathbf{V}) - \mathbb{E}[\mathbf{U} \mid \mathbf{V}]\|^2 + \mathbb{E}\|\mathbf{U} - \mathbb{E}[\mathbf{U} \mid \mathbf{V}]\|^2.$$

*Proof.*  $\mathbf{m}(\mathbf{V}) := \mathbb{E}[\mathbf{U} \mid \mathbf{V}]$ :

$$\begin{aligned} \mathbf{f} - \mathbf{U} &= (\mathbf{f} - \mathbf{m}) + (\mathbf{m} - \mathbf{U}) \xrightarrow{\mathbf{f} - \mathbf{m} \text{ is } \sigma(\mathbf{V})\text{-measurable}} \mathbb{E}[(\mathbf{f} - \mathbf{m}) \cdot (\mathbf{m} - \mathbf{U})] = \mathbb{E}[(\mathbf{f} - \mathbf{m}) \cdot \mathbb{E}[\mathbf{m} - \mathbf{U} \mid \mathbf{V}]] = 0 \\ &\xrightarrow{\text{expand the square}} \mathbb{E}\|\mathbf{f} - \mathbf{U}\|^2 = \mathbb{E}\|\mathbf{f} - \mathbf{m}\|^2 + \mathbb{E}\|\mathbf{U} - \mathbf{m}\|^2. \end{aligned}$$

□

**Theorem 1.2.6** (Flow-matching regression). *For measurable  $\widehat{\mathbf{b}}$  and  $\mathcal{L}_{\text{FM}}(\widehat{\mathbf{b}}) := \mathbb{E}\|\widehat{\mathbf{b}}_\tau(\mathbf{I}_\tau) - \dot{\mathbf{I}}_\tau\|^2$ ,*

$$(1.2.1) \quad \mathcal{L}_{\text{FM}}(\widehat{\mathbf{b}}) = \mathbb{E}\|\widehat{\mathbf{b}}_\tau(\mathbf{I}_\tau) - \mathbf{b}_\tau(\mathbf{I}_\tau)\|^2 + \mathcal{L}_{\text{FM}}(\mathbf{b}),$$

*so  $\mathbf{b}$  minimizes  $\mathcal{L}_{\text{FM}}$ , uniquely in  $L^2(dt \mu_t)$ .*

*Proof.*

$$\begin{aligned} \mathbf{U} := \dot{\mathbf{I}}_\tau, \quad \mathbf{V} := (\tau, \mathbf{I}_\tau) \xrightarrow{\tau \perp (\mathbf{X}_0, \mathbf{X}_1), 1.2.2} \mathbb{E}[\mathbf{U} \mid \mathbf{V}] = \mathbf{b}_\tau(\mathbf{I}_\tau) \xrightarrow{1.2.5} (1.2.1) \\ \xrightarrow{\text{second term free of } \widehat{\mathbf{b}}} \mathbf{b} \in \underset{\widehat{\mathbf{b}}}{\text{argmin}} \mathcal{L}_{\text{FM}}(\widehat{\mathbf{b}}). \end{aligned}$$

□

**Remark 1.2.7** (Simulation-free training). Evaluating  $\mathcal{L}_{\text{FM}}$  needs only samples of  $(\tau, \mathbf{X}_0, \mathbf{X}_1)$ , the value  $\mathbf{I}_\tau$ , and the explicit target  $\dot{\mathbf{I}}_\tau$ ; the learned flow is never simulated during training [1, 14].

**Theorem 1.2.8** (Stability of approximate probability flows). *Let  $\dot{\mathbf{X}}_t = \mathbf{b}_t(\mathbf{X}_t)$ ,  $\widehat{\mathbf{X}}_t = \widehat{\mathbf{b}}_t(\widehat{\mathbf{X}}_t)$ ,  $\mathbf{X}_0 = \widehat{\mathbf{X}}_0$  almost surely,  $\mathbf{b}_t$  uniformly  $L$ -Lipschitz, and  $e_t := \mathbb{E}\|\mathbf{X}_t - \widehat{\mathbf{X}}_t\|^2$ . Then*

$$W_2^2(\text{Law}(\mathbf{X}_t), \text{Law}(\widehat{\mathbf{X}}_t)) \leq e_t \leq e^{(2L+1)t} \int_0^t \mathbb{E}\|\mathbf{b}_s(\widehat{\mathbf{X}}_s) - \widehat{\mathbf{b}}_s(\widehat{\mathbf{X}}_s)\|^2 ds.$$

*Proof.*

$$\begin{aligned} \frac{d}{dt} e_t &= 2\mathbb{E}\left[(\mathbf{X}_t - \widehat{\mathbf{X}}_t) \cdot (\mathbf{b}_t(\mathbf{X}_t) - \widehat{\mathbf{b}}_t(\widehat{\mathbf{X}}_t))\right] \xrightarrow{\text{split at } \mathbf{b}_t(\widehat{\mathbf{X}}_t), \text{Lipschitz, Young}} (2L+1)e_t + \mathbb{E}\|\mathbf{b}_t(\widehat{\mathbf{X}}_t) - \widehat{\mathbf{b}}_t(\widehat{\mathbf{X}}_t)\|^2 \\ &\xrightarrow{e_0=0, \text{Grönwall}} e_t \leq e^{(2L+1)t} \int_0^t \mathbb{E}\|\mathbf{b}_s(\widehat{\mathbf{X}}_s) - \widehat{\mathbf{b}}_s(\widehat{\mathbf{X}}_s)\|^2 ds \xrightarrow{(\mathbf{X}_t, \widehat{\mathbf{X}}_t) \text{ is a coupling}} W_2^2 \leq e_t. \end{aligned}$$

□

**Definition 1.2.9** (One-sided Gaussian interpolant). For  $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_d)$ ,  $\mathbf{X}_1 \sim \mu_1$ ,  $\mathbf{Z} \perp \mathbf{X}_1$ , and schedules with  $\alpha_0 = \beta_1 = 1$ ,  $\alpha_1 = \beta_0 = 0$ ,  $\alpha_t > 0$  on  $[0, 1)$ ,

$$\mathbf{I}_t := \alpha_t \mathbf{Z} + \beta_t \mathbf{X}_1, \quad \eta_t^z(\mathbf{x}) := \mathbb{E}[\mathbf{Z} \mid \mathbf{I}_t = \mathbf{x}], \quad \eta_t^1(\mathbf{x}) := \mathbb{E}[\mathbf{X}_1 \mid \mathbf{I}_t = \mathbf{x}].$$

**Theorem 1.2.10** (Score and denoiser identities). *For  $0 \leq t < 1$ ,*

$$(1.2.2) \quad \mathbf{s}_t(\mathbf{x}) = -\frac{1}{\alpha_t} \eta_t^z(\mathbf{x}) = -\frac{1}{\alpha_t^2} (\mathbf{x} - \beta_t \eta_t^1(\mathbf{x})), \quad \eta_t^1(\mathbf{x}) = \frac{\mathbf{x} + \alpha_t^2 \mathbf{s}_t(\mathbf{x})}{\beta_t} \quad (\beta_t \neq 0).$$

*Proof.*

$$\begin{aligned} p_t(\mathbf{x}) &= \int (2\pi\alpha_t^2)^{-d/2} \exp\left(-\frac{\|\mathbf{x} - \beta_t \mathbf{y}\|^2}{2\alpha_t^2}\right) \mu_1(d\mathbf{y}) \xrightarrow{\nabla_{\mathbf{x}} \text{ under the integral}} \nabla p_t(\mathbf{x}) = -\frac{1}{\alpha_t^2} \int (\mathbf{x} - \beta_t \mathbf{y}) \cdots \mu_1(d\mathbf{y}) \\ &\xrightarrow{\text{Bayes: the integrand is } p_t(\mathbf{x}) \mu_1(d\mathbf{y} \mid \mathbf{I}_t = \mathbf{x})} \mathbf{s}_t = -\frac{\mathbf{x} - \beta_t \eta_t^1}{\alpha_t^2} \xleftrightarrow{\alpha_t \eta_t^z = \mathbf{x} - \beta_t \eta_t^1} (1.2.2). \end{aligned}$$

□

**Proposition 1.2.11** (Velocity through denoisers and score). *For (1.2.9) put  $\gamma_t := \alpha_t \dot{\beta}_t - \dot{\alpha}_t \beta_t$ . Then*

$$\begin{aligned} \mathbf{b}_t &= \dot{\alpha}_t \eta_t^z + \dot{\beta}_t \eta_t^1, \quad \mathbf{s}_t(\mathbf{x}) = -\frac{\dot{\beta}_t \mathbf{x} - \beta_t \mathbf{b}_t(\mathbf{x})}{\alpha_t \gamma_t} \quad (\alpha_t \gamma_t \neq 0), \\ \mathbf{b}_t(\mathbf{x}) &= \frac{\dot{\beta}_t}{\beta_t} \mathbf{x} + \left(\dot{\alpha}_t - \frac{\alpha_t \dot{\beta}_t}{\beta_t}\right) \eta_t^z(\mathbf{x}) \quad (\beta_t \neq 0). \end{aligned}$$

*Proof.*

$$\begin{aligned} \mathbf{x} &= \alpha_t \eta_t^z + \beta_t \eta_t^1, \quad \mathbf{b}_t = \dot{\alpha}_t \eta_t^z + \dot{\beta}_t \eta_t^1 \xrightarrow{\text{eliminate } \eta_t^1} \dot{\beta}_t \mathbf{x} - \beta_t \mathbf{b}_t = \gamma_t \eta_t^z \xrightarrow{1.2.10} \mathbf{s}_t = -\frac{\dot{\beta}_t \mathbf{x} - \beta_t \mathbf{b}_t}{\alpha_t \gamma_t}, \\ \eta_t^1 &= \beta_t^{-1} (\mathbf{x} - \alpha_t \eta_t^z) \xrightarrow{\text{substitute into } \mathbf{b}_t} \mathbf{b}_t = \frac{\dot{\beta}_t}{\beta_t} \mathbf{x} + \left( \dot{\alpha}_t - \frac{\alpha_t \dot{\beta}_t}{\beta_t} \right) \eta_t^z. \end{aligned}$$

□

**Example 1.2.12** (Three schedule families).

| schedule            | $\alpha_t$       | $\beta_t$       | $\gamma_t$         |
|---------------------|------------------|-----------------|--------------------|
| linear              | $1 - t$          | $t$             | $1$                |
| trigonometric       | $\cos(\pi t/2)$  | $\sin(\pi t/2)$ | $\pi/2$            |
| variance preserving | $\sqrt{1 - t^2}$ | $t$             | $(1 - t^2)^{-1/2}$ |

with the last row restricted to  $0 \leq t < 1$ .

**Example 1.2.13** (Gaussian-mixture target). For  $\mu_1 = \sum_{j=1}^J q_j \mathbf{N}(\mathbf{m}_j, C_j)$ ,

$$\begin{aligned} \mathbf{I}_t &\sim \sum_{j=1}^J q_j \mathbf{N}(\beta_t \mathbf{m}_j, C_t^j), \quad C_t^j := \alpha_t^2 I_d + \beta_t^2 C_j, \\ w_j(t, \mathbf{x}) &:= \frac{q_j \det(C_t^j)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \beta_t \mathbf{m}_j)^\top \cdot (C_t^j)^{-1} \cdot (\mathbf{x} - \beta_t \mathbf{m}_j)\right)}{\sum_{\ell=1}^J q_\ell \det(C_t^\ell)^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \beta_t \mathbf{m}_\ell)^\top \cdot (C_t^\ell)^{-1} \cdot (\mathbf{x} - \beta_t \mathbf{m}_\ell)\right)} \\ &\xrightarrow{1.2.10} \mathbf{s}_t(\mathbf{x}) = -\sum_{j=1}^J w_j(t, \mathbf{x}) (C_t^j)^{-1} \cdot (\mathbf{x} - \beta_t \mathbf{m}_j). \end{aligned}$$

**Example 1.2.14** (Centered Gaussian target). For  $\mathbf{X}_1 \sim \mathbf{N}(\mathbf{0}, C)$  and  $C_t = \alpha_t^2 I_d + \beta_t^2 C$ ,

$$\mathbf{s}_t(\mathbf{x}) = -C_t^{-1} \cdot \mathbf{x} \xrightarrow{1.2.11} \mathbf{b}_t(\mathbf{x}) = \frac{1}{2} \dot{C}_t \cdot C_t^{-1} \cdot \mathbf{x} \xrightarrow{C_t, \dot{C}_t \text{ polynomial in } C, \text{ hence commuting}} \mathbf{X}_t = C_t^{1/2} \cdot C_0^{-1/2} \cdot \mathbf{X}_0.$$

**Proposition 1.2.15** (Conditional-mean stepping). *Let  $0 = t_0 < t_1 < \dots < t_K = 1$ ,  $\alpha_k := \alpha_{t_k}$ ,  $\beta_k := \beta_{t_k}$ . Then*

$$\mathbb{E}[\mathbf{I}_s \mid \mathbf{I}_t = \mathbf{x}] = \alpha_s \eta_t^0(\mathbf{x}) + \beta_s \eta_t^1(\mathbf{x}), \quad \mathbf{X}_{k+1} := \alpha_{k+1} \eta_{t_k}^0(\mathbf{X}_k) + \beta_{k+1} \eta_{t_k}^1(\mathbf{X}_k),$$

*and the difference quotient tends to  $\mathbf{b}_{t_k}(\mathbf{X}_k)$  as  $t_{k+1} - t_k \rightarrow 0$ .*

*Proof.*

$$\begin{aligned} \mathbf{I}_s &= \alpha_s \mathbf{X}_0 + \beta_s \mathbf{X}_1 \xrightarrow{\text{linearity, 1.2.2}} \mathbb{E}[\mathbf{I}_s \mid \mathbf{I}_t = \mathbf{x}] = \alpha_s \eta_t^0(\mathbf{x}) + \beta_s \eta_t^1(\mathbf{x}) \\ &\xrightarrow{\text{subtract } \mathbf{X}_k = \alpha_k \eta_{t_k}^0 + \beta_k \eta_{t_k}^1} \frac{\mathbf{X}_{k+1} - \mathbf{X}_k}{t_{k+1} - t_k} = \frac{\alpha_{k+1} - \alpha_k}{t_{k+1} - t_k} \eta_{t_k}^0 + \frac{\beta_{k+1} - \beta_k}{t_{k+1} - t_k} \eta_{t_k}^1 \\ &\xrightarrow{\alpha, \beta \in C^1, 1.2.11} \mathbf{b}_{t_k}(\mathbf{X}_k) \quad (4). \end{aligned}$$

□

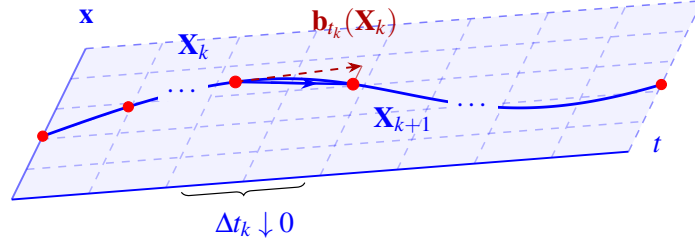


FIGURE 4. Conditional-mean secants recover the probability-flow velocity.

## 2. DIFFUSIVE INTERPOLANTS

### 2.1. Tunable Noise.

**Theorem 2.1.1** (Marginal-preserving diffusion family). *Let  $p_t > 0$  be smooth with  $\partial_t p_t + \nabla \cdot (\mathbf{b}_t p_t) = 0$ , let  $\mathbf{s}_t = \nabla \log p_t$ , and let  $\varepsilon_t \geq 0$ . Then*

$$(2.1.1) \quad \partial_t p_t + \nabla \cdot [(\mathbf{b}_t + \varepsilon_t \mathbf{s}_t) p_t] = \varepsilon_t \Delta p_t,$$

so, whenever the Fokker–Planck equation is unique, the solution of

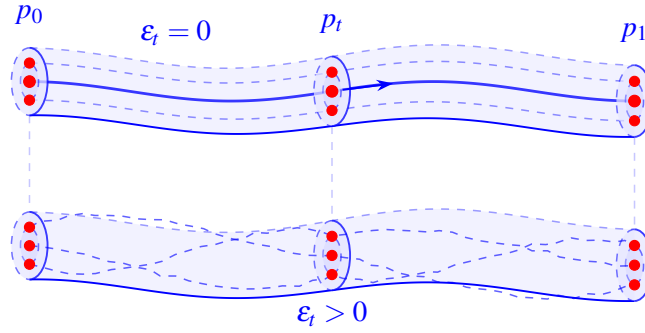
$$d\mathbf{X}_t^\varepsilon = (\mathbf{b}_t + \varepsilon_t \mathbf{s}_t)(\mathbf{X}_t^\varepsilon) dt + \sqrt{2\varepsilon_t} d\mathbf{W}_t, \quad \mathbf{X}_0^\varepsilon \sim \mu_0,$$

has law  $p_t$ , and  $\varepsilon_t = 0$  is the probability-flow equation of (1.2.3).

*Proof.*

$$\begin{aligned} \mathbf{s}_t &= \frac{\nabla p_t}{p_t} \xrightarrow{\nabla \cdot (\mathbf{s}_t p_t) = \nabla \cdot (\nabla p_t)} \nabla \cdot (\mathbf{s}_t p_t) = \Delta p_t \xrightarrow{\partial_t p_t + \nabla \cdot (\mathbf{b}_t p_t) = 0} \Delta p_t \quad (2.1.1) \\ &\xrightarrow{\text{Fokker–Planck of the displayed diffusion, } p_0 = p_0, \text{ uniqueness}} \text{Law}(\mathbf{X}_t^\varepsilon) = p_t \quad (5). \end{aligned}$$

□


 FIGURE 5. Probability flow and score-corrected diffusion share the marginals  $p_t$ .

**Proposition 2.1.2** (Ornstein–Uhlenbeck representation). *Let  $d\mathbf{Y}_\tau = -\mathbf{Y}_\tau d\tau + \sqrt{2} d\mathbf{W}_\tau$  with  $\mathbf{Y}_0 \sim \mu_1$ . Then*

$$\mathbf{Y}_\tau \stackrel{\text{law}}{=} e^{-\tau} \mathbf{X}_1 + \sqrt{1 - e^{-2\tau}} \mathbf{Z}, \quad \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_d), \quad \mathbf{Z} \perp \mathbf{X}_1,$$

and under  $t = e^{-\tau}$  the same marginal path is the interpolant of (1.2.9) with  $\alpha_t = \sqrt{1 - t^2}$ ,  $\beta_t = t$ .

*Proof.*

$$\begin{aligned} d(e^\tau \mathbf{Y}_\tau) &= \sqrt{2} e^\tau d\mathbf{W}_\tau \xrightarrow{\text{integrate}} \mathbf{Y}_\tau = e^{-\tau} \mathbf{Y}_0 + \sqrt{2} \int_0^\tau e^{-(\tau-s)} d\mathbf{W}_s \xrightarrow{\text{Itô isometry: } 2 \int_0^\tau e^{-2(\tau-s)} ds = 1 - e^{-2\tau}} e^{-\tau} \mathbf{X}_1 + \sqrt{1 - e^{-2\tau}} \mathbf{Z} \\ &\xLeftrightarrow{t=e^{-\tau}} \mathbf{I}_t = \sqrt{1 - t^2} \mathbf{Z} + t \mathbf{X}_1 \quad (6). \end{aligned}$$

□

**Corollary 2.1.3** (Variance-preserving probability flow). *For  $\alpha_t = \sqrt{1 - t^2}$ ,  $\beta_t = t$ , and  $0 < t < 1$ ,*

$$(2.1.2) \quad \mathbf{b}_t(\mathbf{x}) = \frac{\mathbf{x}}{t} + \left( \frac{1}{t} - t \right) \mathbf{s}_t(\mathbf{x}), \quad d\mathbf{X}_t = \left[ \frac{\mathbf{X}_t}{t} + \left( \frac{1}{t} - t + \varepsilon_t \right) \mathbf{s}_t(\mathbf{X}_t) \right] dt + \sqrt{2\varepsilon_t} d\mathbf{W}_t.$$

*Proof.*

$$\begin{aligned} \dot{\beta}_t &= 1, \quad \dot{\alpha}_t = -\frac{t}{\sqrt{1-t^2}}, \quad \gamma_t = \frac{1}{\sqrt{1-t^2}} \xrightarrow{1.2.11} \mathbf{x} - t\mathbf{b}_t(\mathbf{x}) = -(1-t^2)\mathbf{s}_t(\mathbf{x}) \xLeftrightarrow{\text{divide by } t} (2.1.2) \\ &\xrightarrow{2.1.1} d\mathbf{X}_t = \left[ \frac{\mathbf{X}_t}{t} + \left( \frac{1}{t} - t + \varepsilon_t \right) \mathbf{s}_t(\mathbf{X}_t) \right] dt + \sqrt{2\varepsilon_t} d\mathbf{W}_t. \end{aligned}$$

□

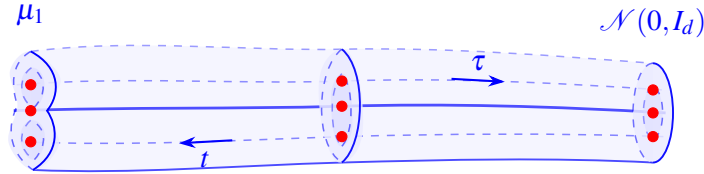


FIGURE 6. Under  $t = e^{-\tau}$ , OU noising and variance-preserving interpolation have identical one-time laws.

## 2.2. Brownian Paths.

**Definition 2.2.1** (Brownian stochastic interpolant). Let  $\mathbf{W} \perp \mathbf{X}_1$  and let  $\sigma, \beta \in C^1([0, 1])$  satisfy  $\beta_0 = 0$ ,  $\beta_1 = 1$ ,  $\sigma_1 = 0$ . Then

$$\mathbf{I}_t^W := \sigma_t \mathbf{W}_t + \beta_t \mathbf{X}_1 \stackrel{\text{law}}{=} \sigma_t \sqrt{t} \mathbf{Z} + \beta_t \mathbf{X}_1 \quad (\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_d)), \quad \mathbf{b}_t := \mathbb{E}[\dot{\sigma}_t \mathbf{W}_t + \dot{\beta}_t \mathbf{X}_1 \mid \mathbf{I}_t^W = \cdot].$$

**Proposition 2.2.2** (Itô evolution and conditional drift). *With  $p_t$  the density of  $\mathbf{I}_t^W$ ,*

$$\begin{aligned} d\mathbf{I}_t^W &= (\dot{\sigma}_t \mathbf{W}_t + \dot{\beta}_t \mathbf{X}_1) dt + \sigma_t d\mathbf{W}_t, \\ (2.2.1) \quad \partial_t p_t + \nabla \cdot (\mathbf{b}_t p_t) &= \frac{1}{2} \sigma_t^2 \Delta p_t. \end{aligned}$$

*Proof.*

$$\begin{aligned} d(\sigma_t \mathbf{W}_t) &= \dot{\sigma}_t \mathbf{W}_t dt + \sigma_t d\mathbf{W}_t, \quad d(\beta_t \mathbf{X}_1) = \dot{\beta}_t \mathbf{X}_1 dt \xrightarrow{\text{Itô on } \varphi \in C_c^2, \text{ then } \mathbb{E}} \frac{d}{dt} \mathbb{E} \varphi(\mathbf{I}_t^W) = \mathbb{E} [\nabla \varphi \cdot \mathbf{b}_t + \frac{1}{2} \sigma_t^2 \Delta \varphi] (\mathbf{I}_t^W) \\ &\xrightarrow{\text{tower over } \sigma(\mathbf{I}_t^W), \text{ integrate by parts}} (2.2.1). \end{aligned}$$

□

**Proposition 2.2.3** (Brownian denoising score). *For  $t > 0$  with  $\sigma_t \neq 0$ ,*

$$\mathbf{s}_t(\mathbf{x}) = -\frac{1}{\sigma_t \sqrt{t}} \mathbb{E}[\mathbf{Z} \mid \mathbf{I}_t^W = \mathbf{x}] = -\frac{1}{\sigma_t t} \mathbb{E}[\mathbf{W}_t \mid \mathbf{I}_t^W = \mathbf{x}].$$

*Proof.*

$$\alpha_t := \sigma_t \sqrt{t} \xrightarrow{2.2.1, 1.2.10} \mathbf{s}_t(\mathbf{x}) = -\frac{1}{\sigma_t \sqrt{t}} \mathbb{E}[\mathbf{Z} \mid \mathbf{I}_t^W = \mathbf{x}] \stackrel{\mathbf{W}_t \stackrel{\text{law}}{=} \sqrt{t} \mathbf{Z} \text{ jointly with the channel}}{\iff} -\frac{1}{\sigma_t t} \mathbb{E}[\mathbf{W}_t \mid \mathbf{I}_t^W = \mathbf{x}].$$

□

**Proposition 2.2.4** (Drift regression). *For  $\tau \sim \text{Unif}(0, 1)$  independent and  $\mathcal{L}_W(\hat{\mathbf{b}}) := \mathbb{E} \|\hat{\mathbf{b}}_\tau(\mathbf{I}_\tau^W) - (\dot{\sigma}_\tau \mathbf{W}_\tau + \dot{\beta}_\tau \mathbf{X}_1)\|^2$ ,*

$$\mathcal{L}_W(\hat{\mathbf{b}}) = \mathbb{E} \|\hat{\mathbf{b}}_\tau(\mathbf{I}_\tau^W) - \mathbf{b}_\tau(\mathbf{I}_\tau^W)\|^2 + \mathcal{L}_W(\mathbf{b}).$$

*Proof.*

$$\begin{aligned} \mathbf{U} &:= \dot{\sigma}_\tau \mathbf{W}_\tau + \dot{\beta}_\tau \mathbf{X}_1, \quad \mathbf{V} := (\tau, \mathbf{I}_\tau^W) \xrightarrow{2.2.1} \mathbb{E}[\mathbf{U} \mid \mathbf{V}] = \mathbf{b}_\tau(\mathbf{I}_\tau^W) \\ &\xrightarrow{1.2.5} \mathcal{L}_W(\hat{\mathbf{b}}) = \mathbb{E} \|\hat{\mathbf{b}}_\tau(\mathbf{I}_\tau^W) - \mathbf{b}_\tau(\mathbf{I}_\tau^W)\|^2 + \mathcal{L}_W(\mathbf{b}). \end{aligned}$$

□

**Theorem 2.2.5** (Equivalent samplers). *Let  $p_t$  be the law of  $\mathbf{I}_t^W$ ,  $\mathbf{s}_t = \nabla \log p_t$ , and  $\varepsilon_t \geq 0$ . Then, whenever the Fokker–Planck equation is unique,*

$$d\mathbf{X}_t^\varepsilon = [\mathbf{b}_t + (\varepsilon_t - \tfrac{1}{2} \sigma_t^2) \mathbf{s}_t](\mathbf{X}_t^\varepsilon) dt + \sqrt{2\varepsilon_t} d\mathbf{W}_t$$

*has law  $p_t$ ;  $\varepsilon_t = \frac{1}{2} \sigma_t^2$  restores the drift  $\mathbf{b}_t$ , and  $\varepsilon_t = 0$  gives  $\dot{\mathbf{X}}_t = (\mathbf{b}_t - \frac{1}{2} \sigma_t^2 \mathbf{s}_t)(\mathbf{X}_t)$ .*

*Proof.*

$$\begin{aligned} \nabla \cdot (\mathbf{s}_t p_t) &= \Delta p_t \xrightarrow{2.2.1} \partial_t p_t + \nabla \cdot [(\mathbf{b}_t + (\varepsilon_t - \tfrac{1}{2} \sigma_t^2) \mathbf{s}_t) p_t] = \varepsilon_t \Delta p_t \\ &\xrightarrow{\text{Fokker–Planck of the displayed diffusion, uniqueness}} \text{Law}(\mathbf{X}_t^\varepsilon) = p_t \quad (7). \end{aligned}$$

□

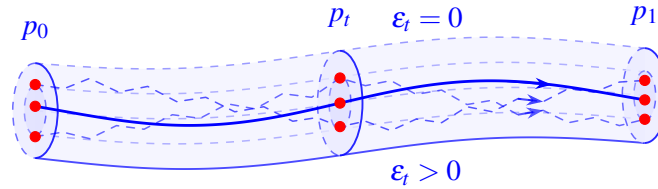


FIGURE 7. The tunable samplers share  $p_t$ , not their path laws.

## Part 2. Lecture B<sup>4</sup>

### 3. LANGEVIN SAMPLING

Throughout, the target is a Gibbs law  $\pi \propto e^{-U}$  accessible only through  $\nabla U$  or through data,  $P$  is a Markov kernel,  $q_t$  is the law of a continuous-time process, and  $\hat{q}_k$  the law of its discretization. Discrepancies are TV,  $W_2$ ,  $D_{\text{KL}}$ , and FI;  $C_{\text{LSI}}$  is a log-Sobolev constant [15, 9, 18, 4].

#### 3.1. Kernels.

**Definition 3.1.1** (Gibbs target and score). For  $U : \mathbb{R}^d \rightarrow \mathbb{R}$  with  $0 < Z := \int_{\mathbb{R}^d} e^{-U} d\mathbf{x} < \infty$ ,

$$\pi(d\mathbf{x}) := Z^{-1} e^{-U(\mathbf{x})} d\mathbf{x}, \quad \nabla \log \pi = -\nabla U,$$

and the task is to produce  $\mathbf{X}$  with  $\text{Law}(\mathbf{X})$  near  $\pi$  without evaluating  $Z$ .

**Definition 3.1.2** (Total variation, Wasserstein distance, relative entropy). For probability measures  $\mu, \nu$  on  $\mathbb{R}^d$ ,

$$\begin{aligned} \text{TV}(\mu, \nu) &:= \sup_A |\mu(A) - \nu(A)|, & W_2^2(\mu, \nu) &:= \inf_{\gamma \in \Pi(\mu, \nu)} \int \|\mathbf{x} - \mathbf{y}\|^2 d\gamma, \\ D_{\text{KL}}(\mu \| \nu) &:= \int \log \left( \frac{d\mu}{d\nu} \right) d\mu \quad (\mu \ll \nu), & D_{\text{KL}}(\mu \| \nu) &:= +\infty \quad \text{otherwise.} \end{aligned}$$

**Proposition 3.1.3** (Pinsker and data processing). For  $\mu \ll \nu$  and measurable  $T$ ,

$$\text{TV}(\mu, \nu) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(\mu \| \nu)}, \quad D_{\text{KL}}(T_{\#}\mu \| T_{\#}\nu) \leq D_{\text{KL}}(\mu \| \nu), \quad \text{TV}(T_{\#}\mu, T_{\#}\nu) \leq \text{TV}(\mu, \nu).$$

*Proof.* [4, §5], [15]. □

**Warning 3.1.4** (Metric and nonmetric error splittings). For a metric  $D$  and an algorithmic invariant law  $\pi_h$ ,

$$D(\text{Law}(\mathbf{X}_n), \pi) \leq \underbrace{D(\text{Law}(\mathbf{X}_n), \pi_h)}_{\text{mixing}} + \underbrace{D(\pi_h, \pi)}_{\text{discretization bias}},$$

and no such triangle inequality is available for  $D_{\text{KL}}$ .

**Definition 3.1.5** (Markov kernel, stationarity, detailed balance). A Markov kernel  $P$  assigns to each  $\mathbf{x}$  a probability measure  $P(\mathbf{x}, \cdot)$ , with

$$(Pf)(\mathbf{x}) := \int f(\mathbf{y}) P(\mathbf{x}, d\mathbf{y}), \quad (\mu P)(A) := \int P(\mathbf{x}, A) \mu(d\mathbf{x}), \quad \text{Law}(\mathbf{X}_n) = \text{Law}(\mathbf{X}_0) P^n.$$

The law  $\pi$  is stationary for  $P$  when  $\pi P = \pi$ , and reversible when

$$\pi(d\mathbf{x}) P(\mathbf{x}, d\mathbf{y}) = \pi(d\mathbf{y}) P(\mathbf{y}, d\mathbf{x})$$

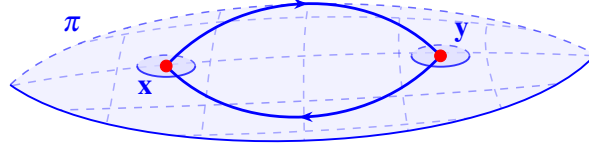
as measures on  $\mathbb{R}^d \times \mathbb{R}^d$ .

**Proposition 3.1.6** (Detailed balance implies stationarity). *Reversibility in (3.1.5) implies  $\pi P = \pi$ .*

*Proof.* Exercise 6.5; see (8). □

---

<sup>4</sup>Based on lectures by Eric Vanden-Eijnden [22].


 FIGURE 8. Symmetry of the equilibrium flux yields  $\pi P = \pi$ .

**Warning 3.1.7** (Stationarity is not convergence). The kernel  $P(\mathbf{x}, \cdot) = \delta_{\mathbf{x}}$  fixes every law and never mixes; the deterministic flip on  $\{0, 1\}$  has stationary law  $\frac{1}{2}(\delta_0 + \delta_1)$  and period two. Convergence needs an ergodicity mechanism.

**Remark 3.1.8** (Design principle).

$\pi \longrightarrow$  continuous-time Markov process with invariant law  $\pi \longrightarrow$  implementable chain,

where the middle stage supplies a generator, a law-level partial differential equation, and a dissipation mechanism, and the last stage introduces a numerical error that may destroy invariance.

### 3.2. Dissipation.

**Definition 3.2.1** (Brownian motion and the Langevin diffusion). Standard  $d$ -dimensional Brownian motion  $(\mathbf{W}_t)_{t \geq 0}$  has continuous paths,  $\mathbf{W}_0 = \mathbf{0}$ , independent increments, and  $\mathbf{W}_t - \mathbf{W}_s \sim N(\mathbf{0}, (t-s)I_d)$ . For  $\pi$  as in (3.1.1),

$$d\mathbf{X}_t = -\nabla U(\mathbf{X}_t) dt + \sqrt{2} d\mathbf{W}_t = \nabla \log \pi(\mathbf{X}_t) dt + \sqrt{2} d\mathbf{W}_t.$$

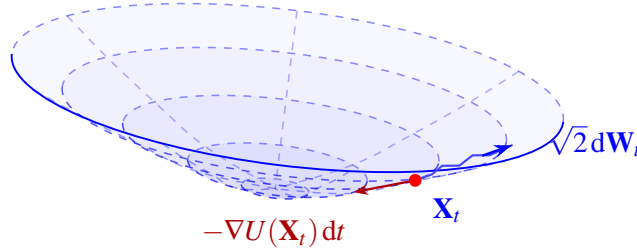


FIGURE 9. Langevin motion combines potential descent with Brownian spreading.

**Proposition 3.2.2** (Itô formula for constant diffusion). For  $d\mathbf{X}_t = \mathbf{b}(\mathbf{X}_t) dt + \sigma d\mathbf{W}_t$  with constant  $\sigma$  and  $\varphi \in C^2(\mathbb{R}^d)$ ,

$$d\varphi(\mathbf{X}_t) = \left[ \nabla \varphi^\top \cdot \mathbf{b} + \frac{1}{2} \text{tr}(\sigma \sigma^\top \cdot \nabla^2 \varphi) \right] (\mathbf{X}_t) dt + \nabla \varphi(\mathbf{X}_t)^\top \cdot \sigma d\mathbf{W}_t.$$

*Proof.* [9, §4], [18, §4]. □

**Definition 3.2.3** (Generator, adjoint, and semigroup). For (3.2.1),

$$\begin{aligned} \mathcal{L}\varphi &:= -\nabla U^\top \cdot \nabla \varphi + \Delta \varphi, & \mathcal{L}^* q &:= \nabla \cdot (q \nabla U) + \Delta q, & (P_t f)(\mathbf{x}) &:= \mathbb{E}[f(\mathbf{X}_t) \mid \mathbf{X}_0 = \mathbf{x}], \\ \partial_t (P_t f) &= \mathcal{L}(P_t f), & P_0 f &= f, & \partial_t q_t &= \mathcal{L}^* q_t. \end{aligned}$$

**Theorem 3.2.4** (Fokker–Planck equation). The density  $q_t$  of  $\mathbf{X}_t$  in (3.2.1) satisfies

$$(3.2.1) \quad \partial_t q_t = \mathcal{L}^* q_t = \nabla \cdot (q_t \nabla U) + \Delta q_t = \nabla \cdot \left( q_t \nabla \log \frac{q_t}{\pi} \right),$$

and  $\mathcal{L}^* \pi = 0$ , hence  $q_0 = \pi$  implies  $q_t = \pi$  for all  $t \geq 0$ .

*Proof.*

$$(3.2.1) \xrightarrow{3.2.2, \mathbb{E}} \frac{d}{dt} \int \phi q_t d\mathbf{x} = \int (\mathcal{L}\phi) q_t d\mathbf{x} \xrightarrow{\text{parts, } \phi \in C_c^\infty \text{ arbitrary}} \partial_t q_t = \nabla \cdot (q_t \nabla U) + \Delta q_t$$

$$\nabla \log \pi = -\nabla U, \text{ so } q_t \nabla U + \nabla q_t = q_t \nabla \log(q_t/\pi) \quad (3.2.1), \quad \nabla \pi = -\pi \nabla U \xrightarrow{3.2.3} \mathcal{L}^* \pi = \nabla \cdot (\pi \nabla U + \nabla \pi) = 0.$$

□

**Example 3.2.5** (Ornstein–Uhlenbeck process). For  $U = \frac{1}{2} \|\mathbf{x}\|^2$  and  $\pi = N(\mathbf{0}, I_d)$ ,

$$d\mathbf{X}_t = -\mathbf{X}_t dt + \sqrt{2} d\mathbf{W}_t, \quad \mathbf{X}_t = e^{-t} \mathbf{X}_0 + \sqrt{2} \int_0^t e^{-(t-s)} d\mathbf{W}_s,$$

$$\mathbf{X}_t \sim N(e^{-t} \mathbf{X}_0, (1 - e^{-2t}) I_d) \xrightarrow[t \rightarrow \infty]{\text{law}} N(\mathbf{0}, I_d).$$

**Definition 3.2.6** (Relative Fisher information and log-Sobolev inequality). For  $q \ll \pi$ ,

$$\text{FI}(q \|\pi) := \int \left\| \nabla \log \frac{q}{\pi} \right\|^2 q d\mathbf{x}, \quad \pi \in \text{LSI}(C_{\text{LSI}}) \xLeftrightarrow{q \ll \pi} D_{\text{KL}}(q \|\pi) \leq \frac{C_{\text{LSI}}}{2} \text{FI}(q \|\pi).$$

**Theorem 3.2.7** (Entropy dissipation and exponential convergence). A regular solution of (3.2.1) satisfies

$$\frac{d}{dt} D_{\text{KL}}(q_t \|\pi) = -\text{FI}(q_t \|\pi),$$

and  $\pi \in \text{LSI}(C_{\text{LSI}})$  gives  $D_{\text{KL}}(q_t \|\pi) \leq e^{-2t/C_{\text{LSI}}} D_{\text{KL}}(q_0 \|\pi)$ .

*Proof.*

$$\begin{aligned} \int \partial_t q_t d\mathbf{x} &= 0 \xrightarrow{\text{differentiate } D_{\text{KL}}} \frac{d}{dt} D_{\text{KL}}(q_t \|\pi) = \int \partial_t q_t \log \frac{q_t}{\pi} d\mathbf{x} \xrightarrow{3.2.1, \text{ parts}} - \int q_t \left\| \nabla \log \frac{q_t}{\pi} \right\|^2 d\mathbf{x} \\ &\xrightarrow{3.2.6} \frac{d}{dt} D_{\text{KL}}(q_t \|\pi) \leq -\frac{2}{C_{\text{LSI}}} D_{\text{KL}}(q_t \|\pi) \xrightarrow{\text{Grönwall}} D_{\text{KL}}(q_t \|\pi) \leq e^{-2t/C_{\text{LSI}}} D_{\text{KL}}(q_0 \|\pi). \end{aligned}$$

□

**Corollary 3.2.8** (Strongly convex potentials). If  $\nabla^2 U \succeq m I_d$  with  $m > 0$ , then  $C_{\text{LSI}} \leq m^{-1}$  and  $D_{\text{KL}}(q_t \|\pi) \leq e^{-2mt} D_{\text{KL}}(q_0 \|\pi)$ .

*Proof.* Bakry–Émery [4, §5.7]; see (10). □

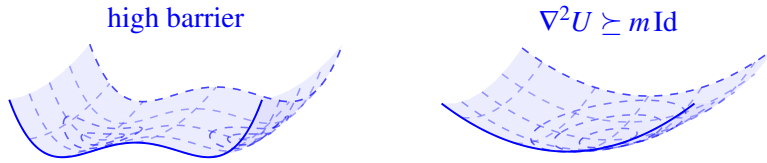


FIGURE 10. High-barrier and uniformly convex landscapes.

### 3.3. Discretization.

**Definition 3.3.1** (Unadjusted Langevin algorithm). For a step size  $h > 0$ ,

$$(3.3.1) \quad \mathbf{X}_{k+1} = \mathbf{X}_k - h\nabla U(\mathbf{X}_k) + \sqrt{2h}\xi_k, \quad \xi_k \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, I_d), \quad Q_h(\mathbf{x}, \cdot) = \mathcal{N}(\mathbf{x} - h\nabla U(\mathbf{x}), 2hI_d).$$

*Warning 3.3.2* (Persistent discretization bias). The kernel (3.3.1) has an invariant law  $\pi_h \neq \pi$ ; for  $D \in \{\text{TV}, W_2\}$  the splitting of (3.1.4) applies, its second term persisting at fixed  $h$ .

**Example 3.3.3** (Exact Gaussian bias). For  $d = 1$ ,  $U(x) = x^2/2$ ,  $\pi = \mathcal{N}(0, 1)$ ,

$$X_{k+1} = (1-h)X_k + \sqrt{2h}\xi_k \xrightarrow{\text{stationary variance}} v_h = (1-h)^2v_h + 2h \xLeftrightarrow{\text{solve}} v_h = \frac{2}{2-h}, \quad \pi_h = \mathcal{N}(0, v_h),$$

$$D_{\text{KL}}(\pi_h \| \pi) = \frac{1}{2}(v_h - 1 - \log v_h), \quad W_2^2(\pi_h, \pi) = (\sqrt{v_h} - 1)^2, \quad v_h = 1 + \frac{h}{2} + \frac{h^2}{4} + O(h^3),$$

so all three quantities equal  $h^2/16 + O(h^3)$ .

**Theorem 3.3.4** (Vempala–Wibisono). Assume that  $\pi \in \text{LSI}(C_{\text{LSI}})$ , that  $\nabla U$  is  $L$ -Lipschitz, that  $0 < h \leq (4C_{\text{LSI}}L^2)^{-1}$ , and that  $D_{\text{KL}}(\hat{q}_0 \| \pi) < \infty$ . Then

$$(3.3.2) \quad D_{\text{KL}}(\hat{q}_k \| \pi) \leq e^{-kh/C_{\text{LSI}}} D_{\text{KL}}(\hat{q}_0 \| \pi) + 8C_{\text{LSI}}L^2dh.$$

*Proof.* Interpolate one step by  $d\bar{\mathbf{X}}_t = -\nabla U(\bar{\mathbf{X}}_0)dt + \sqrt{2}d\mathbf{W}_t$  on  $[0, h]$ ,  $\mathbf{v}_t := \text{Law}(\bar{\mathbf{X}}_t)$ ,  $\mathbf{v}_0 = \hat{q}_k$ ,  $\mathbf{v}_h = \hat{q}_{k+1}$ :

$$\begin{aligned} \frac{d}{dt} D_{\text{KL}}(\mathbf{v}_t \| \pi) &= -\text{FI}(\mathbf{v}_t \| \pi) + \mathbb{E} \left[ (\nabla U(\bar{\mathbf{X}}_t) - \nabla U(\bar{\mathbf{X}}_0))^{\top} \cdot \nabla \log \frac{\mathbf{v}_t}{\pi}(\bar{\mathbf{X}}_t) \right] \\ &\stackrel{\text{Young, } L\text{-smoothness, 3.2.6}}{\implies} \frac{d}{dt} D_{\text{KL}}(\mathbf{v}_t \| \pi) \leq -\frac{1}{C_{\text{LSI}}} D_{\text{KL}}(\mathbf{v}_t \| \pi) + \frac{L^2}{2} \mathbb{E} \|\bar{\mathbf{X}}_t - \bar{\mathbf{X}}_0\|^2, \\ \mathbb{E} \|\bar{\mathbf{X}}_t - \bar{\mathbf{X}}_0\|^2 &= t^2 \mathbb{E}_{\hat{q}_k} \|\nabla U\|^2 + 2dt, \quad \mathbb{E}_{\pi} \|\nabla U\|^2 = \mathbb{E}_{\pi} [\Delta U] \leq Ld, \\ \mathbb{E}_{\hat{q}_k} \|\nabla U\|^2 &\leq 2Ld + 2L^2W_2^2(\hat{q}_k, \pi) \xrightarrow{\text{transport}} W_2^2(\hat{q}_k, \pi) \leq 2C_{\text{LSI}}D_{\text{KL}}(\hat{q}_k \| \pi), \end{aligned}$$

so that, absorbing the  $C_{\text{LSI}}L^4h^3$  term by  $h \leq (4C_{\text{LSI}}L^2)^{-1}$ ,

$$D_{\text{KL}}(\hat{q}_{k+1} \| \pi) \leq e^{-h/C_{\text{LSI}}} D_{\text{KL}}(\hat{q}_k \| \pi) + 6L^2dh^2 \stackrel{\sum_j e^{-jh/C_{\text{LSI}}} \leq 4C_{\text{LSI}}/(3h)}{\implies} (3.3.2);$$

see [23]. □

**Definition 3.3.5** (MALA proposal and acceptance). Draw  $\mathbf{Y} \sim Q_h(\mathbf{x}, \cdot)$  of (3.3.1) and accept with probability

$$a_h(\mathbf{x}, \mathbf{y}) := 1 \wedge \frac{\pi(\mathbf{y})q_h(\mathbf{y}, \mathbf{x})}{\pi(\mathbf{x})q_h(\mathbf{x}, \mathbf{y})},$$

where  $q_h$  is the density of  $Q_h$ ; on rejection set  $\mathbf{X}_{k+1} = \mathbf{X}_k$ .

**Proposition 3.3.6** (Exact stationarity of MALA). The MALA kernel  $P_h^{\text{MALA}}$  is  $\pi$ -reversible, hence  $\pi P_h^{\text{MALA}} = \pi$ .

*Proof.*

$$\begin{aligned} \pi(\mathbf{x})q_h(\mathbf{x}, \mathbf{y})a_h(\mathbf{x}, \mathbf{y}) &\stackrel{3.3.5}{\iff} \min \{ \pi(\mathbf{x})q_h(\mathbf{x}, \mathbf{y}), \pi(\mathbf{y})q_h(\mathbf{y}, \mathbf{x}) \} \stackrel{\text{symmetry of min}}{\iff} \pi(\mathbf{y})q_h(\mathbf{y}, \mathbf{x})a_h(\mathbf{y}, \mathbf{x}), \\ &\text{rejection mass diagonal, hence symmetric} \stackrel{3.1.6}{\implies} \pi P_h^{\text{MALA}} = \pi. \end{aligned}$$

□

*Remark 3.3.7* (Correction versus information access). ULA needs only  $\nabla U$  and keeps a fixed-step bias; MALA removes the bias through the density ratio of (3.3.5), which is unavailable when  $\pi$  is represented only by data [15].

#### 4. SCORE DIFFUSIONS

Throughout, the target is  $p_{\text{data}}$ , the forward process has marginals  $p_t$  with score  $\mathbf{s}_t^* = \nabla \log p_t$ , and  $\mathbf{s}_t$  is a learned surrogate; the Gaussian channel of (1.2.9) reappears with schedule  $(a_t, \sigma_t)$  [12, 20, 15].

##### 4.1. Reversal.

**Definition 4.1.1** (Forward noising and linear Gaussian schedule). For  $\mathbf{X}_0 \sim p_{\text{data}}$ ,

$$d\mathbf{X}_t = \mathbf{f}_t(\mathbf{X}_t) dt + g_t d\mathbf{B}_t \quad (0 \leq t \leq T),$$

$$\mathbf{X}_t \mid \mathbf{X}_0 \sim \mathcal{N}(a_t \mathbf{X}_0, \sigma_t^2 I_d), \quad \mathbf{X}_t \stackrel{\text{law}}{=} a_t \mathbf{X}_0 + \sigma_t \mathbf{Z}, \quad \mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_d).$$

**Example 4.1.2** (Variance-exploding and variance-preserving schedules).

$$d\mathbf{X}_t = d\mathbf{B}_t \implies a_t = 1, \quad \sigma_t^2 = t, \quad p_t = p_{\text{data}} * \mathcal{N}(\mathbf{0}, tI_d),$$

$$d\mathbf{X}_t = -\frac{1}{2}\mathbf{X}_t dt + d\mathbf{B}_t \implies a_t = e^{-t/2}, \quad \sigma_t^2 = 1 - e^{-t}, \quad a_t^2 + \sigma_t^2 = 1.$$

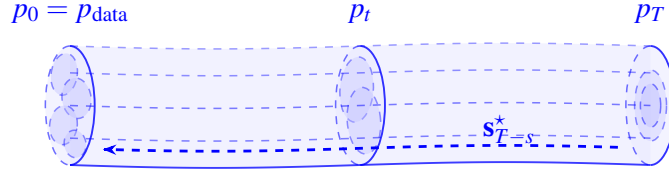


FIGURE 11. Forward noising and exact-score reversal traverse the same marginals.

**Definition 4.1.3** (Time-dependent score and Bayes denoiser). For  $t > 0$  and  $a_t \neq 0$ ,

$$\mathbf{s}_t^*(\mathbf{x}) := \nabla \log p_t(\mathbf{x}), \quad \mathbf{D}_t^*(\mathbf{x}) := \mathbb{E}[\mathbf{X}_0 \mid \mathbf{X}_t = \mathbf{x}].$$

**Theorem 4.1.4** (Continuous-time Tweedie identity). For the channel of (4.1.1),

$$(4.1.1) \quad \mathbf{s}_t^*(\mathbf{x}) = \frac{1}{\sigma_t^2} \mathbb{E}[a_t \mathbf{X}_0 - \mathbf{X}_t \mid \mathbf{X}_t = \mathbf{x}], \quad \mathbf{D}_t^*(\mathbf{x}) = \frac{\mathbf{x} + \sigma_t^2 \mathbf{s}_t^*(\mathbf{x})}{a_t}.$$

*Proof.* With  $\phi_t(\mathbf{x} \mid \mathbf{x}_0) := (2\pi\sigma_t^2)^{-d/2} \exp(-\|\mathbf{x} - a_t \mathbf{x}_0\|^2 / (2\sigma_t^2))$ , differentiation under the integral and Bayes' rule give

$$\nabla_{\mathbf{x}} \phi_t = \frac{a_t \mathbf{x}_0 - \mathbf{x}}{\sigma_t^2} \phi_t \implies \frac{\nabla p_t(\mathbf{x})}{p_t(\mathbf{x})} = \frac{1}{\sigma_t^2} \mathbb{E}[a_t \mathbf{X}_0 - \mathbf{x} \mid \mathbf{X}_t = \mathbf{x}] \implies (4.1.1).$$

□

**Proposition 4.1.5** (Denoising score-matching regression). For  $\mathbf{X}_t = a_t \mathbf{X}_0 + \sigma_t \mathbf{Z}$  with  $\mathbf{Z} \perp \mathbf{X}_0$  and any measurable  $\mathbf{s}_t$ ,

$$\mathbb{E} \left\| \mathbf{s}_t(\mathbf{X}_t) + \frac{\mathbf{Z}}{\sigma_t} \right\|^2 = \mathbb{E} \|\mathbf{s}_t(\mathbf{X}_t) - \mathbf{s}_t^*(\mathbf{X}_t)\|^2 + C_t,$$

with  $C_t$  independent of  $\mathbf{s}_t$ ; the population minimizer is  $\mathbf{s}_t^*$ .

*Proof.*

$$\frac{a_t \mathbf{X}_0 - \mathbf{X}_t}{\sigma_t^2} = -\frac{\mathbf{Z}}{\sigma_t} \xrightarrow{4.1.1} \mathbb{E} \left[ -\frac{\mathbf{Z}}{\sigma_t} \mid \mathbf{X}_t \right] = \mathbf{s}_t^*(\mathbf{X}_t),$$

and (1.2.5) with  $\mathbf{U} = -\mathbf{Z}/\sigma_t$ ,  $\mathbf{V} = \mathbf{X}_t$  gives the decomposition.

□

**Theorem 4.1.6** (Reverse-time diffusion). *Assume the forward dynamics of (4.1.1) has smooth positive marginals. With  $s = T - t$  and  $\mathbf{Y}_0^\leftarrow \sim p_T$ ,*

$$d\mathbf{Y}_s^\leftarrow = [-\mathbf{f}_{T-s} + g_{T-s}^2 \mathbf{s}_{T-s}^*] (\mathbf{Y}_s^\leftarrow) ds + g_{T-s} d\mathbf{B}_s^\leftarrow$$

*satisfies  $\text{Law}(\mathbf{Y}_s^\leftarrow) = p_{T-s}$  for  $0 \leq s \leq T$ .*

*Proof.* With  $q_s := p_{T-s}$  and  $t = T - s$ ,

$$\partial_t p_t = -\nabla \cdot (\mathbf{f}_t p_t) + \frac{1}{2} g_t^2 \Delta p_t \implies \partial_s q_s = \nabla \cdot (\mathbf{f}_t p_t) - \frac{1}{2} g_t^2 \Delta p_t,$$

$$\Delta p_t = \nabla \cdot (p_t \mathbf{s}_t^*) \implies \partial_s q_s = -\nabla \cdot [(-\mathbf{f}_t + g_t^2 \mathbf{s}_t^*) p_t] + \frac{1}{2} g_t^2 \Delta p_t,$$

the Fokker–Planck equation of the reverse dynamics in (4.1.6).  $\square$

**Example 4.1.7** (Reverse dynamics for the two schedules). For (4.1.2),

$$d\mathbf{Y}_s^\leftarrow = \mathbf{s}_{T-s}^* (\mathbf{Y}_s^\leftarrow) ds + d\mathbf{B}_s^\leftarrow, \quad d\mathbf{Y}_s^\leftarrow = [\frac{1}{2} \mathbf{Y}_s^\leftarrow + \mathbf{s}_{T-s}^* (\mathbf{Y}_s^\leftarrow)] ds + d\mathbf{B}_s^\leftarrow.$$

**Warning 4.1.8** (Reversal is not algebraic inversion). Solving  $\mathbf{X}_{k+1} = (1 - h)\mathbf{X}_k + \sqrt{2h}\xi_k$  for  $\mathbf{X}_k$  retains the forward noise and does not give  $\text{Law}(\mathbf{X}_k | \mathbf{X}_{k+1})$ ; time reversal is the Bayes reversal of path measures encoded by (4.1.6).

**Theorem 4.1.9** (Probability flow and its likelihood). *Set  $\mathbf{v}_t := \mathbf{f}_t - \frac{1}{2} g_t^2 \mathbf{s}_t^*$ . If  $\dot{\mathbf{Z}}_t = \mathbf{v}_t(\mathbf{Z}_t)$  and  $\mathbf{Z}_0 \sim p_0$ , then  $\text{Law}(\mathbf{Z}_t) = p_t$ , the reverse flow  $\dot{\mathbf{Z}}_s^\leftarrow = -\mathbf{f}_{T-s} + \frac{1}{2} g_{T-s}^2 \mathbf{s}_{T-s}^*$  has law  $p_{T-s}$ , and*

$$\frac{d}{dt} \log p_t(\mathbf{Z}_t) = -\nabla \cdot \mathbf{v}_t(\mathbf{Z}_t), \quad \log p_0(\mathbf{Z}_0) = \log p_T(\mathbf{Z}_T) + \int_0^T \nabla \cdot \mathbf{v}_t(\mathbf{Z}_t) dt.$$

*Proof.*

$$\Delta p_t = \nabla \cdot (p_t \mathbf{s}_t^*) \implies \partial_t p_t + \nabla \cdot (p_t \mathbf{v}_t) = 0 \xrightarrow{1.1.5} \frac{d}{dt} \log p_t(\mathbf{Z}_t) = -\nabla \cdot \mathbf{v}_t(\mathbf{Z}_t),$$

and reversing time gives the second flow.  $\square$

## 4.2. Localization.

**Definition 4.2.1** (Variance and precision parametrizations). For  $\mathbf{X}_* \sim \mu$  and  $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_d)$  independent,

$$\mathbf{X}_t = \mathbf{X}_* + \sqrt{t} \mathbf{Z}, \quad d\mathbf{Y}_u = \mathbf{X}_* du + d\mathbf{W}_u, \quad \mathbf{Y}_0 = \mathbf{0},$$

$$\mathbf{Y}_u | \mathbf{X}_* = \mathbf{x} \sim \mathcal{N}(u\mathbf{x}, uI_d), \quad \bar{\mathbf{Y}}_u := u^{-1} \mathbf{Y}_u \stackrel{\text{law}}{=} \mathbf{X}_* + \sqrt{t} \mathbf{Z}, \quad t = u^{-1}.$$

**Proposition 4.2.2** (Posterior localization measure). *For  $u > 0$ ,*

$$\mu_u(d\mathbf{x} | \mathbf{y}) = \frac{1}{Z_u(\mathbf{y})} \exp\left(\mathbf{y}^\top \cdot \mathbf{x} - \frac{u}{2} \|\mathbf{x}\|^2\right) \mu(d\mathbf{x}), \quad Z_u(\mathbf{y}) := \int \exp\left(\mathbf{y}^\top \cdot \mathbf{x} - \frac{u}{2} \|\mathbf{x}\|^2\right) \mu(d\mathbf{x}).$$

*Proof.* Expanding the Gaussian exponent of (4.2.1) and applying Bayes' rule,

$$-\frac{\|\mathbf{y} - u\mathbf{x}\|^2}{2u} = -\frac{\|\mathbf{y}\|^2}{2u} + \mathbf{y}^\top \cdot \mathbf{x} - \frac{u}{2} \|\mathbf{x}\|^2 \implies \mu_u(d\mathbf{x} | \mathbf{y}) \propto \exp\left(\mathbf{y}^\top \cdot \mathbf{x} - \frac{u}{2} \|\mathbf{x}\|^2\right) \mu(d\mathbf{x}) \quad (12).$$

$\square$

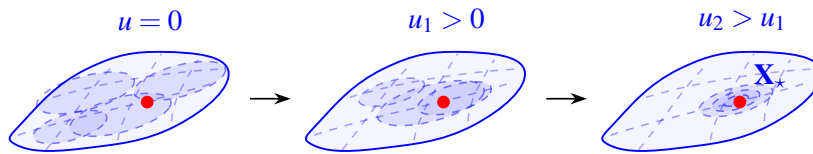


FIGURE 12. Increasing precision localizes the posterior toward  $\mathbf{X}_*$ .

**Definition 4.2.3** (Posterior process, innovation, and moments). With  $\mathcal{F}_u^{\mathbf{Y}} := \sigma(\mathbf{Y}_r : 0 \leq r \leq u)$ ,

$$\begin{aligned} \mu_u &:= \text{Law}(\mathbf{X}_* \mid \mathcal{F}_u^{\mathbf{Y}}), & \mathbf{M}_u &:= \mathbb{E}[\mathbf{X}_* \mid \mathcal{F}_u^{\mathbf{Y}}], & d\mathbf{I}_u &:= d\mathbf{Y}_u - \mathbf{M}_u du, \\ \mathbf{m}_u(\mathbf{y}) &:= \mathbb{E}[\mathbf{X}_* \mid \mathbf{Y}_u = \mathbf{y}], & \Sigma_u(\mathbf{y}) &:= \text{Cov}(\mathbf{X}_* \mid \mathbf{Y}_u = \mathbf{y}), & \Sigma_u &:= \Sigma_u(\mathbf{Y}_u). \end{aligned}$$

**Proposition 4.2.4** (The innovation is Brownian).  $(\mathbf{I}_u)_{u \geq 0}$  is a standard  $(\mathcal{F}_u^{\mathbf{Y}})$ -Brownian motion.

*Proof.* By Lévy's characterization, it suffices that

$$d\mathbf{I}_u = (\mathbf{X}_* - \mathbf{M}_u) du + d\mathbf{W}_u, \quad \mathbb{E}[\mathbf{X}_* - \mathbf{M}_u \mid \mathcal{F}_u^{\mathbf{Y}}] = \mathbf{0}, \quad d[\mathbf{I}]_u = I_d du.$$

□

**Theorem 4.2.5** (Measure-valued localization equation). For bounded measurable  $f$ ,

$$(4.2.1) \quad d \int f d\mu_u = \left[ \int f(\mathbf{x})(\mathbf{x} - \mathbf{M}_u) \mu_u(d\mathbf{x}) \right]^\top \cdot d\mathbf{I}_u, \quad d\mu_u(d\mathbf{x}) = (\mathbf{x} - \mathbf{M}_u)^\top \cdot d\mathbf{I}_u \mu_u(d\mathbf{x}),$$

so  $u \mapsto \mu_u(A)$  is a martingale for every Borel  $A$ .

*Proof.* With  $r_u(\mathbf{x}) := d\mu_u/d\mu = Z_u^{-1} \exp(\mathbf{Y}_u^\top \cdot \mathbf{x} - \frac{u}{2} \|\mathbf{x}\|^2)$  from (4.2.2), Itô's formula applied twice gives

$$\begin{aligned} d \exp \left( \mathbf{Y}_u^\top \cdot \mathbf{x} - \frac{u}{2} \|\mathbf{x}\|^2 \right) &= \exp \left( \mathbf{Y}_u^\top \cdot \mathbf{x} - \frac{u}{2} \|\mathbf{x}\|^2 \right) \mathbf{x}^\top \cdot d\mathbf{Y}_u \implies \frac{dZ_u}{Z_u} = \mathbf{M}_u^\top \cdot d\mathbf{Y}_u, \\ \frac{dr_u(\mathbf{x})}{r_u(\mathbf{x})} &= (\mathbf{x} - \mathbf{M}_u)^\top \cdot (d\mathbf{Y}_u - \mathbf{M}_u du) = (\mathbf{x} - \mathbf{M}_u)^\top \cdot d\mathbf{I}_u \implies (4.2.1). \end{aligned}$$

□

**Proposition 4.2.6** (Localization form of Tweedie and linear response). With  $p_t$  the density of  $\mathbf{X}_* + \sqrt{t}\mathbf{Z}$  and  $t = u^{-1}$ ,

$$\mathbf{m}_u(\mathbf{y}) = \frac{\mathbf{y}}{u} + \frac{1}{u} \nabla \log p_{1/u} \left( \frac{\mathbf{y}}{u} \right), \quad \nabla_{\mathbf{y}} \mathbf{m}_u(\mathbf{y}) = \Sigma_u(\mathbf{y}), \quad \nabla \mathbf{s}_t^*(\mathbf{x}) = \frac{1}{t^2} \Sigma_{1/t} \left( \frac{\mathbf{x}}{t} \right) - \frac{1}{t} I_d.$$

*Proof.* Differentiating the tilt of (4.2.2) in  $\mathbf{y}$ ,

$$\bar{\mathbf{Y}}_u = \mathbf{X}_* + u^{-1/2} \mathbf{Z} \xrightarrow{4.1.1} \mathbf{m}_u(\mathbf{y}) = \frac{\mathbf{y}}{u} + \frac{1}{u} \nabla \log p_{1/u}(\mathbf{y}/u), \quad \partial_{y_j} m_{u,i} = \text{Cov}_u^{\mathbf{y}}(X_i, X_j) = (\Sigma_u)_{ij},$$

and differentiating  $\mathbf{m}_{1/t}(\mathbf{x}/t) = \mathbf{x} + t\mathbf{s}_t^*(\mathbf{x})$  in  $\mathbf{x}$ ,

$$\frac{1}{t} \Sigma_{1/t}(\mathbf{x}/t) = I_d + t \nabla \mathbf{s}_t^*(\mathbf{x}) \implies \nabla \mathbf{s}_t^*(\mathbf{x}) = \frac{1}{t^2} \Sigma_{1/t} \left( \frac{\mathbf{x}}{t} \right) - \frac{1}{t} I_d.$$

□

**Proposition 4.2.7** (Posterior-mean martingale and covariance dissipation). Assume  $\mathbb{E} \|\mathbf{X}_*\|^4 < \infty$ . Then

$$d\mathbf{M}_u = \Sigma_u \cdot d\mathbf{I}_u, \quad \frac{d}{du} \mathbb{E}[\text{tr}(\Sigma_u)] = -\mathbb{E}[\text{tr}(\Sigma_u^2)] \leq 0, \quad \int_0^\infty \mathbb{E}[\text{tr}(\Sigma_u^2)] du \leq \text{tr}(\text{Cov}(\mathbf{X}_*)).$$

In particular  $\text{Cov}(\mathbf{X}_*) \preceq I_d$  gives the bound  $d$ .

*Proof.*

$$f(\mathbf{x}) = \mathbf{x} \xrightarrow{4.2.1} d\mathbf{M}_u = \left[ \int \mathbf{x}(\mathbf{x} - \mathbf{M}_u)^\top d\mu_u \right] \cdot d\mathbf{I}_u = \Sigma_u \cdot d\mathbf{I}_u,$$

and  $f(\mathbf{x}) = \mathbf{x}\mathbf{x}^\top$  with Itô's formula, expectation, and integration over  $[0, \infty)$  give

$$d \text{tr}(\Sigma_u) = \text{martingale} - \text{tr}(\Sigma_u^2) du \implies \frac{d}{du} \mathbb{E}[\text{tr}(\Sigma_u)] = -\mathbb{E}[\text{tr}(\Sigma_u^2)] \implies \int_0^\infty \mathbb{E}[\text{tr}(\Sigma_u^2)] du \leq \text{tr}(\text{Cov}(\mathbf{X}_*)).$$

□

**Theorem 4.2.8** (Finite-dimensional Polchinski equation). *For  $p_t = p_0 * \mathbf{N}(\mathbf{0}, tI_d)$ ,  $U_t := -\log p_t$ , and  $\mathbf{s}_t^* = -\nabla U_t$ ,*

$$\partial_t p_t = \frac{1}{2} \Delta p_t, \quad \partial_t U_t = \frac{1}{2} \Delta U_t - \frac{1}{2} \|\nabla U_t\|^2, \quad \partial_t \mathbf{s}_t^* = \frac{1}{2} \Delta \mathbf{s}_t^* + \nabla \left( \frac{1}{2} \|\mathbf{s}_t^*\|^2 \right).$$

*Proof.* Since  $p_t = e^{-U_t}$  solves the heat equation  $\partial_t p_t = \frac{1}{2} \Delta p_t$ ,

$$\begin{aligned} \Delta p_t = e^{-U_t} (\|\nabla U_t\|^2 - \Delta U_t) &\implies \partial_t U_t = \frac{1}{2} \Delta U_t - \frac{1}{2} \|\nabla U_t\|^2, \\ \mathbf{s}_t^* = -\nabla U_t &\implies \partial_t \mathbf{s}_t^* = -\nabla \partial_t U_t = \frac{1}{2} \Delta \mathbf{s}_t^* + \nabla \left( \frac{1}{2} \|\mathbf{s}_t^*\|^2 \right), \end{aligned}$$

the viscous Burgers form [19, 5]. □

**Remark 4.2.9** (Three languages for one Gaussian channel).

$$\left[ \begin{array}{ll} \text{diffusion} & \mathbf{X}_t = \mathbf{X}_0 + \sqrt{t} \mathbf{Z} \\ \text{localization} & d\mathbf{Y}_u = \mathbf{X}_* du + d\mathbf{W}_u, \quad u = t^{-1} \\ \text{coarse-graining} & p_t = p_0 * \mathbf{N}(\mathbf{0}, tI_d), \quad U_t = -\log p_t \end{array} \right], \quad \mathbf{x} + t\mathbf{s}_t^*(\mathbf{x}) = \mathbf{m}_{1/t}(\mathbf{x}/t)$$

[10, 16].

### 4.3. Bookkeeping.

**Definition 4.3.1** (Forward grid and Gaussian reverse update). Fix  $0 < t_1 = \delta < \dots < t_K = T$ ,  $h_k := t_{k+1} - t_k$ ,  $\mathbf{X}_k := \mathbf{X}_{t_k}$ ,  $p_k := p_{t_k}$ . Then

$$\mathbf{X}_{k+1} = \mathbf{X}_k + \sqrt{h_k} \xi_k, \quad P_k^{\rightarrow}(\mathbf{x} \rightarrow \mathbf{x}') = (2\pi h_k)^{-d/2} \exp\left(-\frac{\|\mathbf{x}' - \mathbf{x}\|^2}{2h_k}\right),$$

and, for a learned score  $\mathbf{s}_{k+1}$ ,

$$\hat{P}_k^{\leftarrow}(\mathbf{x}' \rightarrow \cdot) := \mathbf{N}(\mathbf{x}' + h_k \mathbf{s}_{k+1}(\mathbf{x}'), h_k I_d).$$

**Proposition 4.3.2** (Exact Bayes reverse kernel).

$$P_k^{\leftarrow}(\mathbf{x}' \rightarrow \mathbf{x}) = \frac{p_k(\mathbf{x}) P_k^{\rightarrow}(\mathbf{x} \rightarrow \mathbf{x}')}{p_{k+1}(\mathbf{x}')}, \quad \mathbb{E}[\mathbf{X}_k \mid \mathbf{X}_{k+1} = \mathbf{x}'] = \mathbf{x}' + h_k \mathbf{s}_{k+1}^*(\mathbf{x}').$$

*Proof.* Bayes' rule gives the kernel; with  $a = 1$  and  $\sigma^2 = h_k$  in the one-step channel,

$$\mathbf{X}_{k+1} = \mathbf{X}_k + \sqrt{h_k} \xi_k \xrightarrow{4.1.1} \mathbb{E}[\mathbf{X}_k \mid \mathbf{X}_{k+1} = \mathbf{x}'] = \mathbf{x}' + h_k \mathbf{s}_{k+1}^*(\mathbf{x}') \quad (13).$$

□

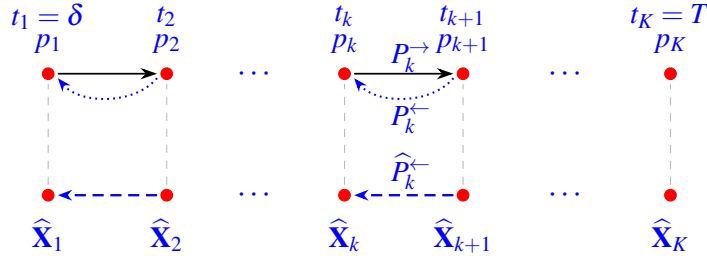


FIGURE 13. The learned reverse recursion approximates the Bayes chain from  $p_T$  to  $p_\delta$ .

**Proposition 4.3.3** (Early-stopping coupling). *For  $\mathbf{X}_1 = a_1 \mathbf{X}_0 + \sigma_1 \mathbf{Z}$  with  $\mathbf{X}_0 \sim p_{\text{data}}$  and  $\mathbf{Z} \perp \mathbf{X}_0$ ,*

$$W_2^2(p_{\text{data}}, p_1) \leq (1 - a_1)^2 \mathbb{E} \|\mathbf{X}_0\|^2 + \sigma_1^2 d, \quad W_2^2(p_{\text{data}}, p_\delta) \leq \delta d \text{ for the variance-exploding grid.}$$

*Proof.*

$\mathbf{X}_1 - \mathbf{X}_0 = (a_1 - 1)\mathbf{X}_0 + \sigma_1 \mathbf{Z}$ ,  $\mathbf{Z} \perp \mathbf{X}_0$ ,  $\mathbb{E}\|\mathbf{Z}\|^2 = d \implies \mathbb{E}\|\mathbf{X}_1 - \mathbf{X}_0\|^2 = (1 - a_1)^2 \mathbb{E}\|\mathbf{X}_0\|^2 + \sigma_1^2 d$ , and  $W_2$  infimizes over couplings.  $\square$

**Definition 4.3.4** (The four error sources). Let  $\hat{p}_K$  be the implemented initialization and  $\hat{p}_1$  the output of the approximate reverse chain. Set

$$D_{\text{KL}}(p_K \|\hat{p}_K) \leq \varepsilon_{\text{init}}^2, \quad \varepsilon_{k,\text{score}}^2 := \mathbb{E}_{p_k} \|\mathbf{s}_k - \mathbf{s}_k^*\|^2,$$

with  $\varepsilon_{\text{disc}}^2$  the residual accumulated when the exact score is used but the reverse kernels are replaced by local Gaussians:

$$\left[ \begin{array}{ll} \text{early stopping} & p_{\text{data}} \leftrightarrow p_1 \\ \text{initialization} & p_K \leftrightarrow \hat{p}_K \\ \text{score estimation} & \mathbf{s}_k^* \leftrightarrow \mathbf{s}_k \\ \text{discretization} & P_k^{\leftarrow} \leftrightarrow \text{local Gaussian step} \end{array} \right].$$

**Lemma 4.3.5** (KL chain rule for reverse Markov paths). For path laws  $\mathbb{P} = \mu_K P_{K-1} \cdots P_1$  and  $\mathbb{Q} = \nu_K Q_{K-1} \cdots Q_1$  with  $\mathbb{P}$ -marginals  $\mu_{k+1}$ ,

$$D_{\text{KL}}(\mathbb{P} \|\mathbb{Q}) = D_{\text{KL}}(\mu_K \|\nu_K) + \sum_{k=1}^{K-1} \mathbb{E}_{\mathbf{x}_{k+1} \sim \mu_{k+1}} \left[ D_{\text{KL}}(P_k(\mathbf{x}_{k+1} \rightarrow \cdot) \| Q_k(\mathbf{x}_{k+1} \rightarrow \cdot)) \right].$$

*Proof.*

$$\log \frac{d\mathbb{P}}{d\mathbb{Q}} = \log \frac{\mu_K}{\nu_K}(\mathbf{x}_K) + \sum_{k=1}^{K-1} \log \frac{P_k(\mathbf{x}_{k+1} \rightarrow \mathbf{x}_k)}{Q_k(\mathbf{x}_{k+1} \rightarrow \mathbf{x}_k)},$$

and taking  $\mathbb{E}_{\mathbb{P}}$  term by term, conditioning the  $k$ -th summand on  $\mathbf{x}_{k+1}$ , gives the claim.  $\square$

**Lemma 4.3.6** (Equal-covariance Gaussian score cost). For  $G_k^*, G_k$  the Gaussian steps of (4.3.1) built from  $\mathbf{s}_{k+1}^*$  and  $\mathbf{s}_{k+1}$ ,

$$D_{\text{KL}}(G_k^*(\mathbf{x}' \rightarrow \cdot) \| G_k(\mathbf{x}' \rightarrow \cdot)) = \frac{h_k}{2} \|\mathbf{s}_{k+1}^*(\mathbf{x}') - \mathbf{s}_{k+1}(\mathbf{x}')\|^2, \quad \mathbb{E}_{p_{k+1}} D_{\text{KL}}(G_k^* \| G_k) = \frac{h_k}{2} \varepsilon_{k+1,\text{score}}^2.$$

*Proof.*

$D_{\text{KL}}(\mathcal{N}(\mathbf{m}, \Sigma) \| \mathcal{N}(\hat{\mathbf{m}}, \Sigma)) = \frac{1}{2}(\mathbf{m} - \hat{\mathbf{m}})^\top \cdot \Sigma^{-1} \cdot (\mathbf{m} - \hat{\mathbf{m}})$ ,  $\mathbf{m} - \hat{\mathbf{m}} = h_k(\mathbf{s}_{k+1}^* - \mathbf{s}_{k+1})$ ,  $\Sigma^{-1} = h_k^{-1} I_d$ , and averaging over  $\mathbf{x}' \sim p_{k+1}$  gives the second identity.  $\square$

**Theorem 4.3.7** (Diffusion-sampler error decomposition). If  $D_{\text{KL}}(p_K \|\hat{p}_K) \leq \varepsilon_{\text{init}}^2$  and the summed local reverse-kernel divergences are at most  $\varepsilon_{\text{disc}}^2 + \frac{1}{2} \sum_k h_k \varepsilon_{k+1,\text{score}}^2$ , then

$$(4.3.1) \quad D_{\text{KL}}(p_1 \|\hat{p}_1) \leq \varepsilon_{\text{init}}^2 + \varepsilon_{\text{disc}}^2 + \frac{1}{2} \sum_{k=1}^{K-1} h_k \varepsilon_{k+1,\text{score}}^2,$$

which together with (4.3.3) separates the raw-data comparison from the reverse-chain error.

*Proof.*

$$T(\mathbf{x}_K, \dots, \mathbf{x}_1) := \mathbf{x}_1, \quad T_{\#} \mathbb{P} = p_1, \quad T_{\#} \hat{\mathbb{P}} = \hat{p}_1 \xrightarrow{3.1.3} D_{\text{KL}}(p_1 \|\hat{p}_1) \leq D_{\text{KL}}(\mathbb{P} \|\hat{\mathbb{P}}) \xrightarrow{4.3.5, 4.3.6} (4.3.1). \quad \square$$

### Part 3. Exercises

#### 5. EXERCISES

**Exercise 5.1** (1.2.3). Construct

$$\mu_0 \xrightarrow{\mathbf{b}} \mu_1, \quad \mu_0 \xrightarrow{\tilde{\mathbf{b}}} \mu_1, \quad (\mu_t)_{t \in [0,1]} \neq (\tilde{\mu}_t)_{t \in [0,1]},$$

and explain why this does not contradict Theorem 1.2.3.

**Exercise 5.2** (1.1.8). Determine equality in

$$\mathbb{E} \|\mathbf{X}_1 - \mathbf{X}_0\|^2 \leq \int_0^1 \int_{\mathbb{R}^d} \|\mathbf{b}_t(\mathbf{x})\|^2 \mu_t(d\mathbf{x}) dt,$$

and relate the equality value to  $\mathcal{C}(\Phi_{1,0})$  of Definition 1.1.7.

**Exercise 5.3** (1.2.4). For

$$\mu_0 = N(\mathbf{0}, I_d), \quad \mu_1 = \frac{1}{2}(\delta_{\mathbf{m}} + \delta_{-\mathbf{m}}),$$

compute  $\mathbf{b}_t$  and its limit as  $t \uparrow 1$ .

**Exercise 5.4** (1.2.6). Show that  $\mathcal{L}_{\text{FM}}^\pi(\mathbf{b})$  depends on the coupling  $\pi$ , and compare independent and deterministic couplings.

**Exercise 5.5** (1.2.6). Using Lemma 1.2.5, show

$$\argmin_{\hat{\mathbf{b}}} \mathbb{E} \|\hat{\mathbf{b}}_\tau(\mathbf{I}_\tau) - \hat{\mathbf{I}}_\tau\|^2 = \argmin_{\hat{\mathbf{b}}} \mathbb{E} \|\hat{\mathbf{b}}_\tau(\mathbf{I}_\tau) - \mathbb{E}[\hat{\mathbf{I}}_\tau | \mathbf{X}_1]\|^2.$$

**Exercise 5.6** (1.2.12). For a reparametrization  $\theta$ , show

$$\tilde{\mu}_t = \mu_{\theta(t)}, \quad \tilde{\mathbf{b}}_t = \dot{\theta}(t) \mathbf{b}_{\theta(t)}, \quad \tilde{\gamma}_t = \dot{\theta}(t) \gamma_{\theta(t)}.$$

**Exercise 5.7** (1.2.13). For  $J = 2$ ,  $C_1 = C_2 = I_d$ , determine

$$\lim_{\|\mathbf{m}_1 - \mathbf{m}_2\| \rightarrow \infty} w_j(t, \mathbf{x}), \quad \lim_{\|\mathbf{m}_1 - \mathbf{m}_2\| \rightarrow \infty} \mathbf{s}_t(\mathbf{x}).$$

**Exercise 5.8** (1.2.15). Show that Proposition 1.2.15 is exact at every  $t_k$  for Gaussian  $\mu_1$ , and exhibit a target for which it fails.

**Exercise 5.9** (2.1.3). Under  $t = e^{-\tau}$ , derive (2.1.2) from Proposition 2.1.2.

**Exercise 5.10** (2.2.5). Determine how  $\varepsilon_t$  trades discretization error against score error, including

$$\varepsilon_t = 0, \quad \varepsilon_t = \frac{1}{2} \sigma_t^2.$$

**Exercise 5.11** (3.1.2). For

$$q_A(a_A | C) := \mathbb{P}(X_{0,A} = a_A | C), \quad \tilde{q}_A := \prod_{i \in A} \mathbb{P}(X_{0,i} = a_i | C),$$

show

$$D_{\text{KL}}(q_A \| \tilde{q}_A) = \sum_{i \in A} H(X_{0,i} | C) - H(X_{0,A} | C) \geq 0.$$

$$q_A = \tilde{q}_A \iff (X_{0,i})_{i \in A} \text{ are conditionally independent given } C.$$

**Exercise 5.12** (3.2.7). For  $q_0 = N(\mathbf{m}_0, C_0)$ , compute

$$D_{\text{KL}}(N(e^{-t} \mathbf{m}_0, I_d + e^{-2t}(C_0 - I_d)) \| N(\mathbf{0}, I_d)),$$

and compare its decay with Definition 3.2.6.

**Exercise 5.13** (3.3.3). For  $A = A^\top > 0$ , show

$$U(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \cdot A \cdot \mathbf{x} \implies \pi_h = N\left(\mathbf{0}, \left(A - \frac{h}{2} A^2\right)^{-1}\right), \quad 0 < h < \frac{2}{\lambda_{\max}(A)}.$$

**Exercise 5.14** (3.3.4). For  $D_0 := D_{\text{KL}}(\hat{q}_0 \parallel \pi)$ , minimize  $k$  subject to

$$e^{-kh/C_{\text{LSI}}} D_0 + 8C_{\text{LSI}} L^2 dh \leq \varepsilon^2, \quad 0 < h \leq (4C_{\text{LSI}} L^2)^{-1},$$

and give the dependence on  $d, L, C_{\text{LSI}}, \varepsilon$ .

**Exercise 5.15** (3.3.5). For  $U(x) = x^2/2$ , compute  $a_h(x, y)$  and determine its behavior as  $h \downarrow 0$  and, for the product target, as  $d \rightarrow \infty$ .

**Exercise 5.16** (4.1.4). For

$$p_{\text{data}} = \sum_j q_j N(\mathbf{m}_j, C_j), \quad C_{j,t} := a_t^2 C_j + \sigma_t^2 I_d,$$

compute the mixture weights and  $\mathbf{s}_t^*$ , and verify (4.1.1).

**Exercise 5.17** (4.2.7). For  $\mu = \frac{1}{2}(\delta_{\mathbf{m}} + \delta_{-\mathbf{m}})$ , show

$$\mathbf{m}_u(\mathbf{y}) = \mathbf{m} \tanh(\mathbf{y}^\top \cdot \mathbf{m}), \quad \Sigma_u(\mathbf{y}) = \mathbf{m} \mathbf{m}^\top \text{sech}^2(\mathbf{y}^\top \cdot \mathbf{m}),$$

and verify Proposition 4.2.7.

**Exercise 5.18** (4.2.8). Show

$$\partial_t U_t = \frac{1}{2} \Delta U_t - \frac{1}{2} \|\nabla U_t\|^2 \stackrel{p_t \propto e^{-U_t}}{\iff} \partial_t p_t = \frac{1}{2} \Delta p_t, \quad p_0 = N(\mathbf{m}, C) \implies \mathbf{s}_t^*(\mathbf{x}) = -(C + tI_d)^{-1} \cdot (\mathbf{x} - \mathbf{m}).$$

**Exercise 5.19** (4.3.7). For fixed  $K$  and  $\varepsilon_{k, \text{score}}^2 \leq \varepsilon^2$ , minimize the right-hand side of (4.3.1) over the grids of Definition 4.3.1, and compare the minimizer with a geometric grid.

## 6. HOMEWORK

**Exercise 6.1** (1.1.4). For

$$\dot{\mathbf{X}}_t = A \cdot \mathbf{X}_t, \quad \mathbf{X}_0 \sim N(\mathbf{0}, C_0),$$

show

$$C_t = e^{tA} \cdot C_0 \cdot e^{tA^\top}, \quad \mu_t = N(\mathbf{0}, C_t), \quad \dot{C}_t = A \cdot C_t + C_t \cdot A^\top, \quad \nabla \cdot \mathbf{b}_t = \text{tr}(A). \\ \partial_t p_t + \nabla \cdot (\mathbf{b}_t p_t) = 0, \quad \log p_1(\mathbf{X}_1) = \log p_0(\mathbf{X}_0) - \text{tr}(A).$$

*Proof.* The linear flow determines the covariance and divergence explicitly.

$$\Phi_{t,0} = e^{tA}, \quad \mathbf{X}_t = e^{tA} \cdot \mathbf{X}_0 \implies C_t = e^{tA} \cdot C_0 \cdot e^{tA^\top}, \quad \mu_t = N(\mathbf{0}, C_t). \\ \dot{C}_t = A \cdot C_t + C_t \cdot A^\top, \quad \mathbf{b}_t(\mathbf{x}) = A \cdot \mathbf{x}, \quad \nabla \cdot \mathbf{b}_t = \text{tr}(A). \\ \partial_t \log p_t = -\frac{1}{2} \text{tr}(C_t^{-1} \cdot \dot{C}_t) + \frac{1}{2} \mathbf{x}^\top \cdot C_t^{-1} \cdot \dot{C}_t \cdot C_t^{-1} \cdot \mathbf{x} = -\text{tr}(A) + \mathbf{x}^\top \cdot C_t^{-1} \cdot A \cdot \mathbf{x} = -\nabla \cdot (\mathbf{b}_t p_t) / p_t. \\ \nabla \cdot \mathbf{b}_t = \text{tr}(A) \stackrel{1.1.5}{\implies} \log p_1(\mathbf{X}_1) = \log p_0(\mathbf{X}_0) - \int_0^1 \text{tr}(A) dt = \log p_0(\mathbf{X}_0) - \text{tr}(A).$$

□

**Exercise 6.2** (1.1.6). For  $\mathbf{v} \sim N(\mathbf{0}, I_d)$ , show

$$\mathbb{E}[\mathbf{v}^\top \cdot D\mathbf{b}_t \cdot \mathbf{v}] = \nabla \cdot \mathbf{b}_t, \quad \text{Var}(\mathbf{v}^\top \cdot D\mathbf{b}_t \cdot \mathbf{v}) = \|D\mathbf{b}_t\|_F^2 + \text{tr}((D\mathbf{b}_t)^2),$$

and determine the variance of the  $m$ -probe estimator.

*Proof.* Apply the Gaussian second- and fourth-moment identities to a matrix  $M$ .

$$\mathbb{E}[v_i v_j] = \delta_{ij} \implies \mathbb{E}[\mathbf{v}^\top \cdot M \cdot \mathbf{v}] = \sum_{i,j} M_{ij} \mathbb{E}[v_i v_j] = \sum_i M_{ii} = \text{tr}(M).$$

$$\mathbb{E}[v_i v_j v_k v_l] = \delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk} \implies \text{Var}(\mathbf{v}^\top \cdot M \cdot \mathbf{v}) = \|M\|_F^2 + \text{tr}(M^2).$$

$$M = D\mathbf{b}_t(\mathbf{x}), \quad \hat{d}_m := \frac{1}{m} \sum_{r=1}^m (\mathbf{v}^{(r)})^\top \cdot M \cdot \mathbf{v}^{(r)} \implies \mathbb{E}[\hat{d}_m] = \nabla \cdot \mathbf{b}_t, \quad \text{Var}(\hat{d}_m) = \frac{\|M\|_F^2 + \text{tr}(M^2)}{m}.$$

Thus  $m$  probes divide the variance by  $m$ , while exact divergence requires  $d$  coordinate traces.  $\square$

**Exercise 6.3** (1.2.8). For  $0 < t \leq 1$ , show

$$e_t \leq \lambda^{-1} e^{(2L+\lambda)t} \int_0^t \mathbb{E}\|r_s\|^2 ds, \quad \inf_{\lambda>0} \lambda^{-1} e^{(2L+\lambda)t} = e t e^{2Lt} \leq e^{(2L+1)t}.$$

Here

$$r_s := \mathbf{b}_s(\hat{\mathbf{X}}_s) - \hat{\mathbf{b}}_s(\hat{\mathbf{X}}_s).$$

*Proof.* Set

$$\begin{aligned} \delta_t &:= \mathbf{X}_t - \hat{\mathbf{X}}_t, & r_t &:= \mathbf{b}_t(\hat{\mathbf{X}}_t) - \hat{\mathbf{b}}_t(\hat{\mathbf{X}}_t), & e_t &:= \mathbb{E}\|\delta_t\|^2. \\ \dot{e}_t &= 2\mathbb{E}[\delta_t^\top \cdot (\mathbf{b}_t(\mathbf{X}_t) - \mathbf{b}_t(\hat{\mathbf{X}}_t))] + 2\mathbb{E}[\delta_t^\top \cdot r_t] \leq (2L + \lambda)e_t + \lambda^{-1} \mathbb{E}\|r_t\|^2. \\ e_0 = 0 &\xrightarrow{\text{Grönwall}} e_t \leq \lambda^{-1} e^{(2L+\lambda)t} \int_0^t \mathbb{E}\|r_s\|^2 ds, & \frac{d}{d\lambda} \log(\lambda^{-1} e^{\lambda t}) &= t - \lambda^{-1}. \\ t - \lambda^{-1} = 0 &\iff \lambda = t^{-1} \implies \inf_{\lambda>0} \lambda^{-1} e^{(2L+\lambda)t} = e t e^{2Lt} \leq e^{(2L+1)t}, & 1 + \log t &\leq t. \end{aligned}$$

$\square$

**Exercise 6.4** (1.2.10). For

$$\mu_1 = N(\mathbf{m}, C), \quad M_t := \alpha_t^2 I_d + \beta_t^2 C,$$

show

$$\mathbf{s}_t(\mathbf{x}) = -M_t^{-1} \cdot (\mathbf{x} - \beta_t \mathbf{m}), \quad \eta_t^1(\mathbf{x}) = M_t^{-1} \cdot (\beta_t C \cdot \mathbf{x} + \alpha_t^2 \mathbf{m}),$$

and recover Example 1.2.14 when  $\mathbf{m} = \mathbf{0}$ .

*Proof.* The interpolant is Gaussian, so its score is affine.

$$\begin{aligned} \mathbf{X}_t &= \alpha_t \mathbf{Z} + \beta_t \mathbf{X}_1, & \mathbf{Z} \perp \mathbf{X}_1 &\implies \mu_t = N(\beta_t \mathbf{m}, M_t), & \mathbf{s}_t(\mathbf{x}) &= -M_t^{-1} \cdot (\mathbf{x} - \beta_t \mathbf{m}). \\ \eta_t^1(\mathbf{x}) &\xrightarrow{1.2.10} \frac{\mathbf{x} + \alpha_t^2 \mathbf{s}_t(\mathbf{x})}{\beta_t} = M_t^{-1} \cdot (\beta_t C \cdot \mathbf{x} + \alpha_t^2 \mathbf{m}), & \mathbf{m} = \mathbf{0} &\implies \eta_t^1(\mathbf{x}) = \beta_t M_t^{-1} C \cdot \mathbf{x}. \\ \mathbf{m} = \mathbf{0} &\implies \mathbf{s}_t(\mathbf{x}) = -M_t^{-1} \cdot \mathbf{x} \xrightarrow{1.2.11} \mathbf{b}_t(\mathbf{x}) = \frac{1}{2} \dot{M}_t \cdot M_t^{-1} \cdot \mathbf{x} \xrightarrow{[M_t, \dot{M}_t]=0} \mathbf{X}_t = M_t^{1/2} \cdot M_0^{-1/2} \cdot \mathbf{X}_0. \end{aligned}$$

$\square$

**Exercise 6.5** (3.1.6). Show

$$\pi(d\mathbf{x})P(\mathbf{x}, d\mathbf{y}) = \pi(d\mathbf{y})P(\mathbf{y}, d\mathbf{x}) \implies \pi P = \pi.$$

*Proof.* Let  $A$  be measurable and integrate detailed balance over  $\mathbf{x}$ .

$$(\pi P)(A) = \iint \mathbf{1}_A(\mathbf{y}) \pi(d\mathbf{x}) P(\mathbf{x}, d\mathbf{y}) \xrightarrow{3.1.5} \iint \mathbf{1}_A(\mathbf{y}) \pi(d\mathbf{y}) P(\mathbf{y}, d\mathbf{x}) = \int \mathbf{1}_A(\mathbf{y}) \pi(d\mathbf{y}) = \pi(A).$$

$\square$

**Exercise 6.6** (3.1.3). Find  $\mu_\varepsilon, \nu$  such that

$$\frac{2\text{TV}(\mu_\varepsilon, \nu)^2}{D_{\text{KL}}(\mu_\varepsilon \| \nu)} \longrightarrow 1,$$

and  $\mu, \nu$  such that

$$\text{TV}(\mu, \nu) < 1, \quad D_{\text{KL}}(\mu \| \nu) = +\infty.$$

*Proof.* Bernoulli laws give both examples.

$$(\mu_\varepsilon, \nu) = (\text{Ber}(\tfrac{1}{2} + \varepsilon), \text{Ber}(\tfrac{1}{2})) \implies \text{TV}(\mu_\varepsilon, \nu) = \varepsilon, \quad D_{\text{KL}}(\mu_\varepsilon \| \nu) = 2\varepsilon^2 + O(\varepsilon^4).$$

$$\frac{2\varepsilon^2}{2\varepsilon^2 + O(\varepsilon^4)} \longrightarrow 1, \quad (\mu, \nu) = (\text{Ber}(\tfrac{1}{2}), \delta_0) \implies \text{TV} = \tfrac{1}{2} < 1, \quad \mu \not\ll \nu \implies D_{\text{KL}} = +\infty.$$

□

**Exercise 6.7** (3.2.3). For

$$d\mathbf{X}_t = \mathbf{b}(\mathbf{X}_t) dt + \sigma(\mathbf{X}_t) d\mathbf{W}_t,$$

show

$$\begin{aligned} \mathcal{L}\varphi &= \mathbf{b}^\top \cdot \nabla \varphi + \tfrac{1}{2} \text{tr}(\sigma \sigma^\top \cdot \nabla^2 \varphi), \quad \pi \text{ invariant} \iff \mathcal{L}^* \pi = 0. \\ \mathcal{L}^* q &= -\nabla \cdot (\mathbf{b}q) + \tfrac{1}{2} \sum_{i,j} \partial_i \partial_j ((\sigma \sigma^\top)_{ij} q). \end{aligned}$$

*Proof.* Set  $a := \sigma \sigma^\top$  and apply Itô's formula.

$$d\varphi(\mathbf{X}_t) = \left( \mathbf{b}^\top \cdot \nabla \varphi + \tfrac{1}{2} \text{tr}(a \cdot \nabla^2 \varphi) \right) dt + (\nabla \varphi)^\top \cdot \sigma d\mathbf{W}_t \implies \mathcal{L}\varphi = \mathbf{b}^\top \cdot \nabla \varphi + \tfrac{1}{2} \text{tr}(a \cdot \nabla^2 \varphi).$$

$$\int (\mathcal{L}\varphi)q = \int \varphi \left[ -\nabla \cdot (\mathbf{b}q) + \tfrac{1}{2} \sum_{i,j} \partial_i \partial_j (a_{ij}q) \right] \implies \mathcal{L}^* q = -\nabla \cdot (\mathbf{b}q) + \tfrac{1}{2} \sum_{i,j} \partial_i \partial_j (a_{ij}q).$$

$$\pi \text{ invariant} \implies \int \mathcal{L}\varphi d\pi = 0 \iff \mathcal{L}^* \pi = 0, \quad \pi \text{ invariant} \iff \mathcal{L}^* \pi = 0.$$

□

**Exercise 6.8** (4.1.9). For  $p_{\text{data}} = \mathbf{N}(\mathbf{0}, C)$ , show

$$\text{Law}(\mathbf{Z}_s^\leftarrow) = \text{Law}(\mathbf{Y}_s^\leftarrow) = \mathbf{N}(\mathbf{0}, C_{T-s}), \quad [\mathbf{Z}^\leftarrow]_s = 0, \quad [\mathbf{Y}^\leftarrow]_s = \left( \int_0^s g_{T-u}^2 du \right) I_d.$$

*Proof.* The common Gaussian score determines both marginal evolutions.

$$p_t = \mathbf{N}(\mathbf{0}, C_t) \implies \mathbf{s}_t^*(\mathbf{x}) = -C_t^{-1} \cdot \mathbf{x} \xrightarrow{4.1.6, 4.1.9} \text{Law}(\mathbf{Z}_s^\leftarrow) = \text{Law}(\mathbf{Y}_s^\leftarrow) = p_{T-s}.$$

$$[\mathbf{Z}^\leftarrow]_T = 0, \quad [\mathbf{Y}^\leftarrow]_T = \left( \int_0^T g_t^2 dt \right) I_d \neq 0 \quad \left( \int_0^T g_t^2 dt > 0 \right) \implies \text{Law}(\mathbf{Z}^\leftarrow) \perp \text{Law}(\mathbf{Y}^\leftarrow).$$

□

**Exercise 6.9** (4.3.3). For  $\mathbf{m} \neq \mathbf{0}$ , show

$$p_{\text{data}} := \tfrac{1}{2}(\delta_{\mathbf{m}} + \delta_{-\mathbf{m}}) \implies D_{\text{KL}}(p_{\text{data}} \| \hat{p}_1) = +\infty \quad \text{for every Gaussian } \hat{p}_1,$$

while the coupling of Proposition 4.3.3 gives

$$W_2^2(p_{\text{data}}, p_1) \leq (1 - a_1)^2 \|\mathbf{m}\|^2 + \sigma_1^2 d < \infty.$$

*Proof.* The two-point law is singular with respect to every Gaussian.

$$\begin{aligned} \hat{p}_1 &:= N(\mathbf{a}, C), \quad C \neq 0 \implies \hat{p}_1(\{\mathbf{m}, -\mathbf{m}\}) = 0, \quad C = 0 \implies \hat{p}_1 = \delta_{\mathbf{a}}. \\ p_{\text{data}} \not\ll \hat{p}_1 &\implies D_{\text{KL}}(p_{\text{data}} \parallel \hat{p}_1) = +\infty \quad \text{for every Gaussian } \hat{p}_1. \\ \mathbf{X}_0 \sim p_{\text{data}}, \quad \mathbf{Z} \sim N(\mathbf{0}, I_d) \perp \mathbf{X}_0, \quad \mathbf{X}_1 = a_1 \mathbf{X}_0 + \sigma_1 \mathbf{Z} &\implies W_2^2(p_{\text{data}}, p_1) \leq \mathbb{E} \|\mathbf{X}_1 - \mathbf{X}_0\|^2. \\ \mathbb{E} \|\mathbf{X}_1 - \mathbf{X}_0\|^2 = (1 - a_1)^2 \|\mathbf{m}\|^2 + \sigma_1^2 d < \infty, \quad a_1 = 1, \quad \sigma_1^2 = \delta &\implies W_2^2(p_{\text{data}}, p_\delta) \leq \delta d. \end{aligned}$$

□

## REFERENCES

- [1] M. S. Albergo and E. Vanden-Eijnden, *Building normalizing flows with stochastic interpolants*, International Conference on Learning Representations, 2023.
- [2] M. S. Albergo, N. M. Boffi, and E. Vanden-Eijnden, *Stochastic interpolants: a unifying framework for flows and diffusions*, J. Mach. Learn. Res. **26** (2025), no. 209, 1–80.
- [3] F. Bach, *Learning Theory from First Principles*, Adaptive Computation and Machine Learning, MIT Press, Cambridge, MA, 2024.
- [4] D. Bakry, I. Gentil, and M. Ledoux, *Analysis and Geometry of Markov Diffusion Operators*, Grundlehren der mathematischen Wissenschaften **348**, Springer, Cham, 2014.
- [5] R. Bauerschmidt, T. Bodineau, and B. Dagallier, *Stochastic dynamics and the Polchinski equation: an introduction*, Probab. Surv. **21** (2024), 200–290.
- [6] J.-D. Benamou and Y. Brenier, *A computational fluid mechanics solution to the Monge–Kantorovich mass transfer problem*, Numer. Math. **84** (2000), 375–393.
- [7] J. Benton, V. De Bortoli, A. Doucet, and G. Deligiannidis, *Nearly  $d$ -linear convergence bounds for diffusion models via stochastic localization*, International Conference on Learning Representations, 2024.
- [8] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud, *Neural ordinary differential equations*, Adv. Neural Inf. Process. Syst. **31** (2018).
- [9] W. E, T. Li, and E. Vanden-Eijnden, *Applied Stochastic Analysis*, Graduate Studies in Mathematics **199**, American Mathematical Society, Providence, RI, 2019.
- [10] R. Eldan, *Thin shell implies spectral gap up to polylog via a stochastic localization scheme*, Geom. Funct. Anal. **23** (2013), 532–569.
- [11] D. J. Higham, *An algorithmic introduction to numerical simulation of stochastic differential equations*, SIAM Rev. **43** (2001), no. 3, 525–546.
- [12] J. Ho, A. Jain, and P. Abbeel, *Denoising diffusion probabilistic models*, Adv. Neural Inf. Process. Syst. **33** (2020), 6840–6851.
- [13] P. E. Kloeden and E. Platen, *Numerical Solution of Stochastic Differential Equations*, Applications of Mathematics **23**, Springer, Berlin, 1999, 3rd corrected printing.
- [14] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, *Flow matching for generative modeling*, International Conference on Learning Representations, 2023.
- [15] J. Lu, *Lecture A. Lecture notes*, Mathematics of Generative Modeling.
- [16] A. Montanari, *Sampling, diffusions, and stochastic localization*, 2023, arXiv:2305.10690.
- [17] K. P. Murphy, *Probabilistic Machine Learning: Advanced Topics*, Adaptive Computation and Machine Learning, MIT Press, Cambridge, MA, 2023.
- [18] B. Øksendal, *Stochastic Differential Equations: An Introduction with Applications*, 6th ed., Universitext, Springer, Berlin, 2003.
- [19] J. Polchinski, *Renormalization and effective Lagrangians*, Nuclear Phys. B **231** (1984), 269–295.
- [20] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, *Score-based generative modeling through stochastic differential equations*, International Conference on Learning Representations, 2021.
- [21] J. M. Tomczak, *Deep Generative Modeling*, 2nd ed., Springer, Cham, 2024.
- [22] E. Vanden-Eijnden, *Lecture B. Lecture notes*, Mathematics of Generative Modeling.
- [23] S. S. Vempala and A. Wibisono, *Rapid convergence of the unadjusted Langevin algorithm: isoperimetry suffices*, Adv. Neural Inf. Process. Syst. **32** (2019).